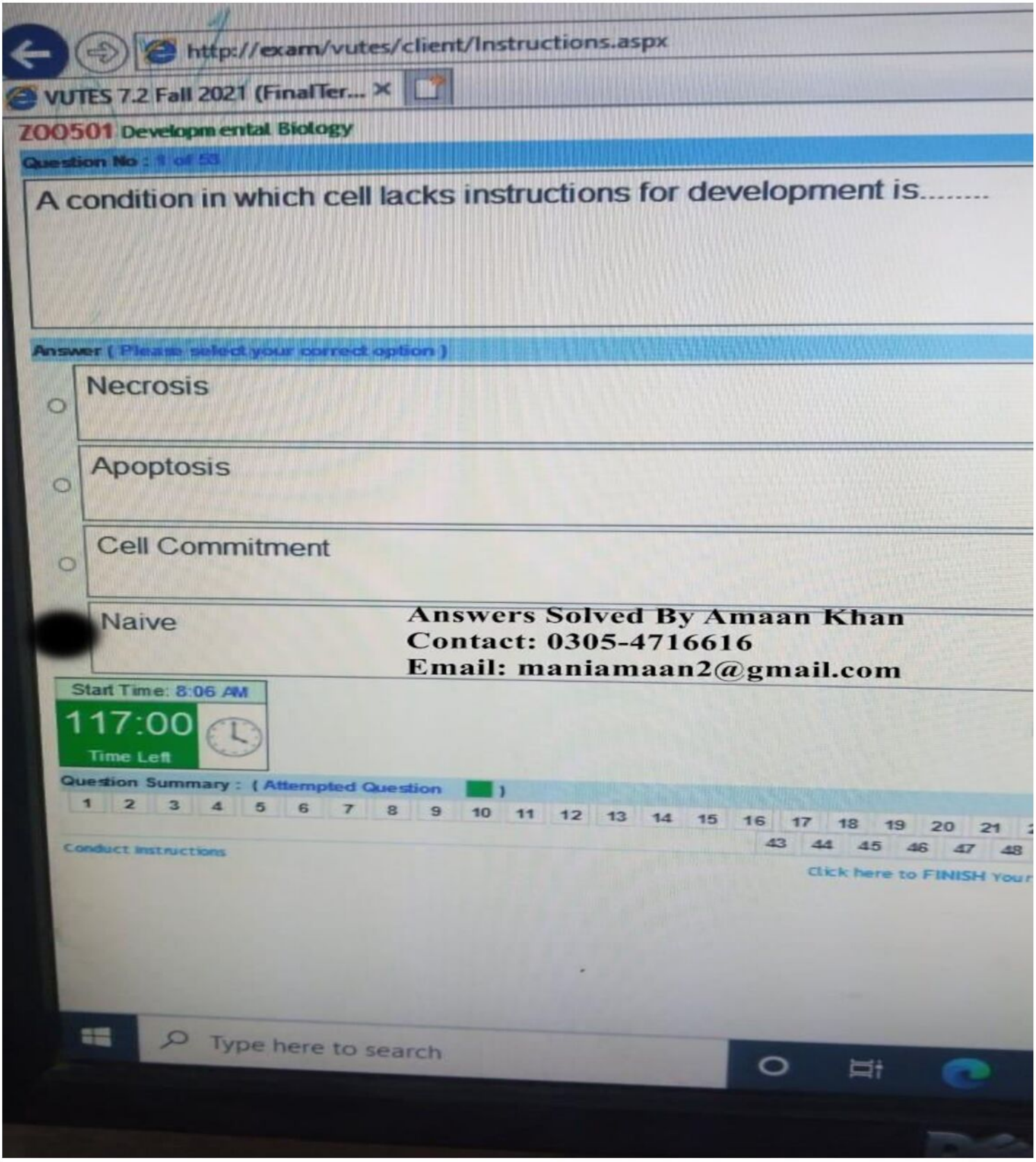
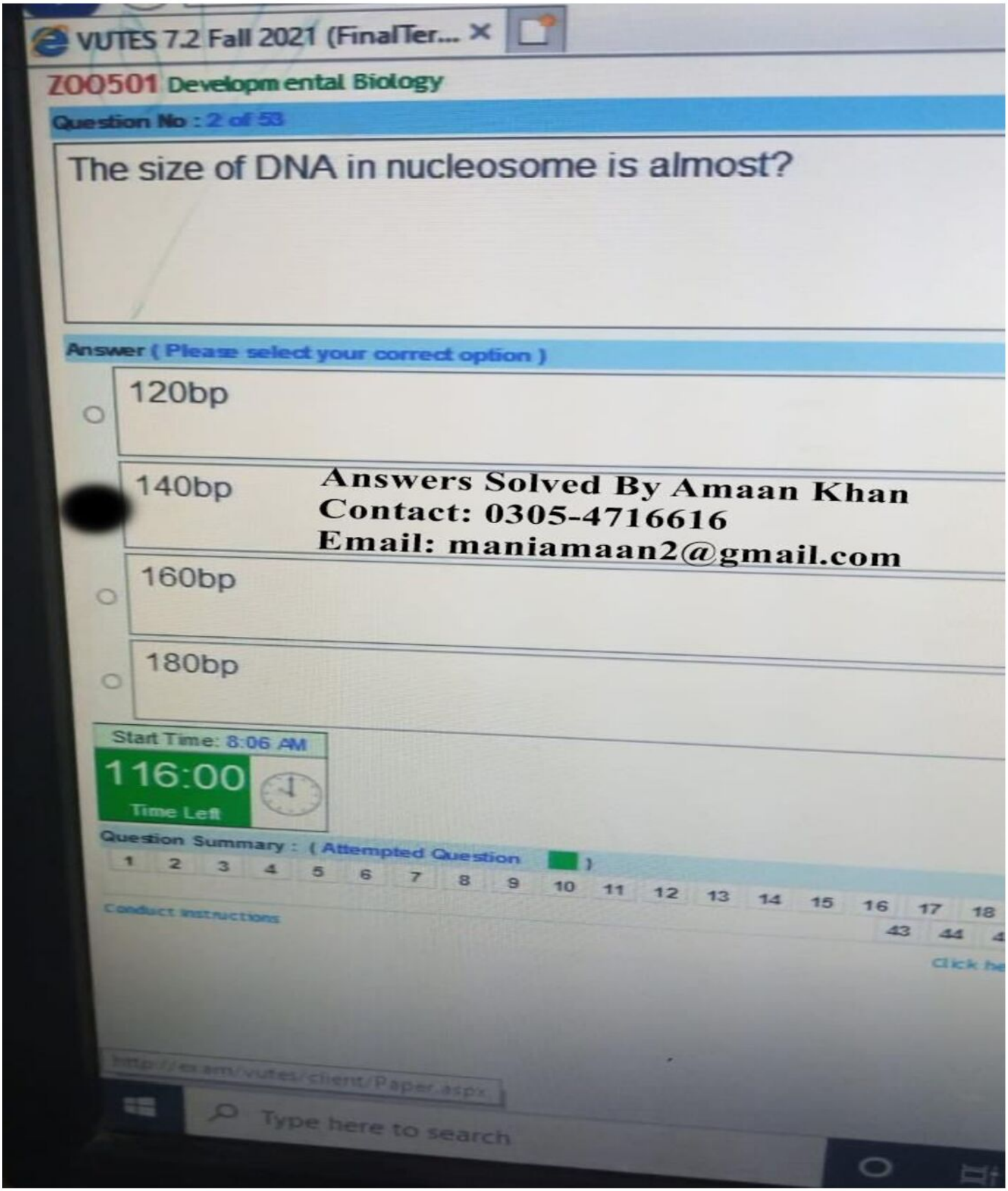
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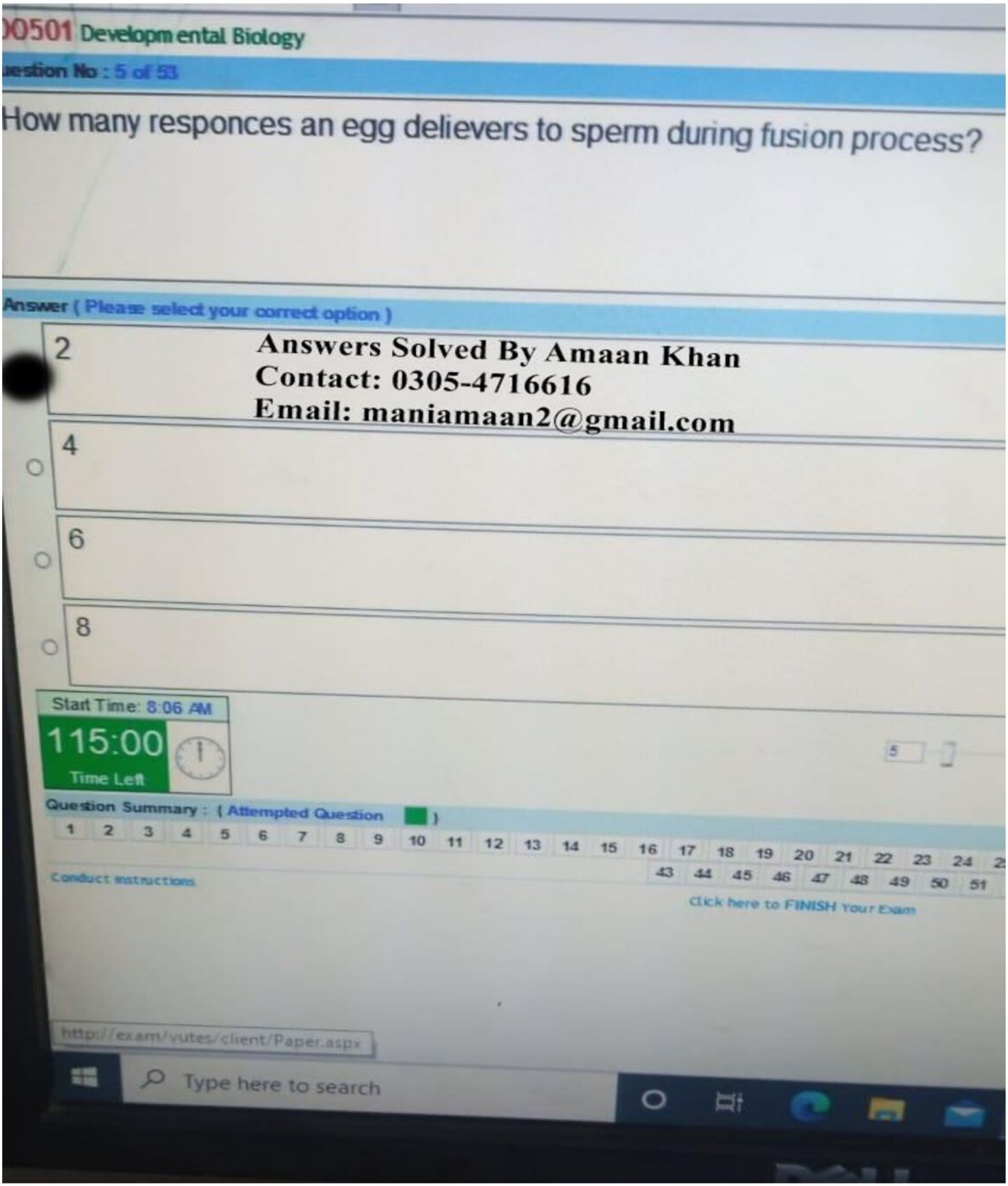
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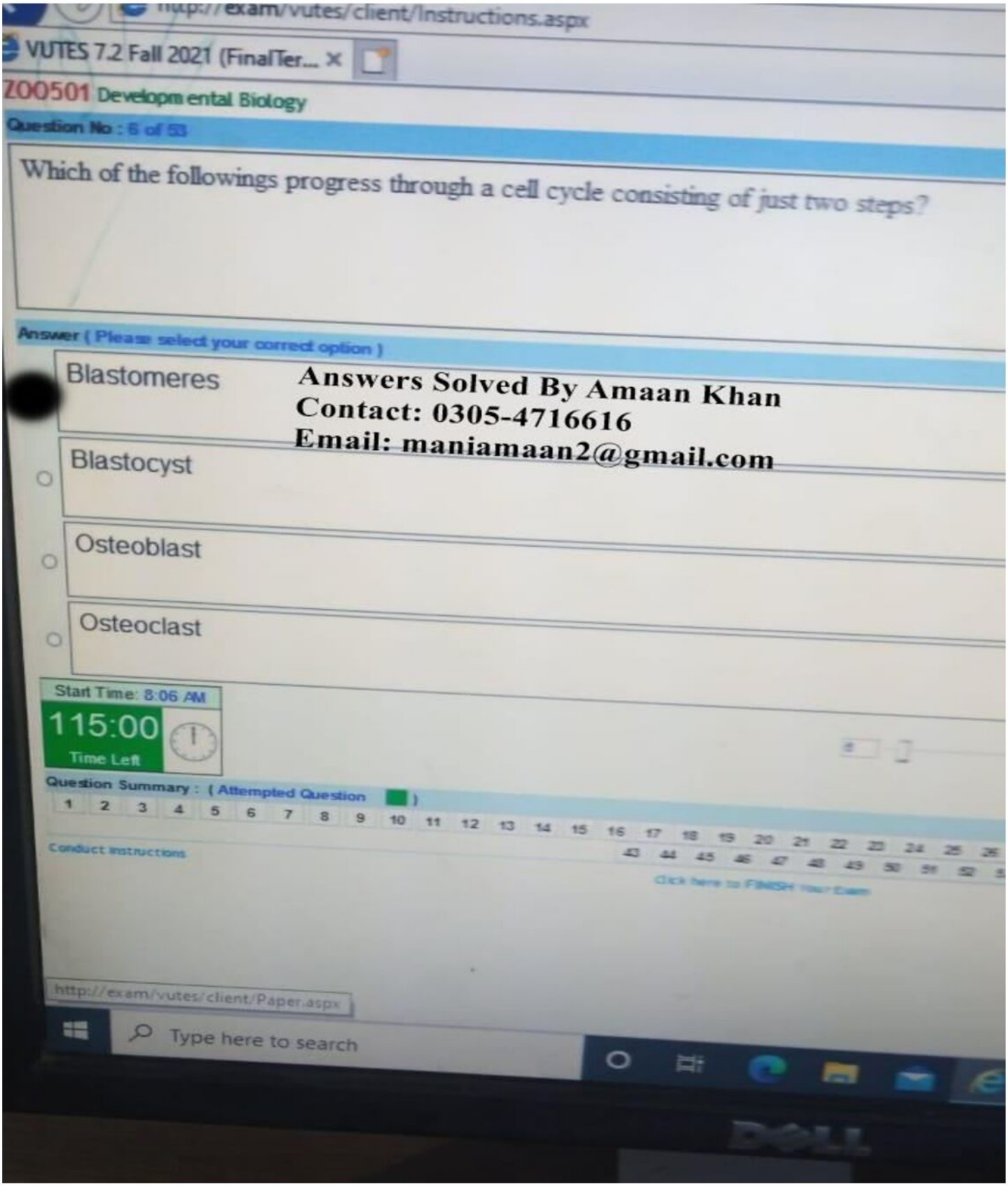
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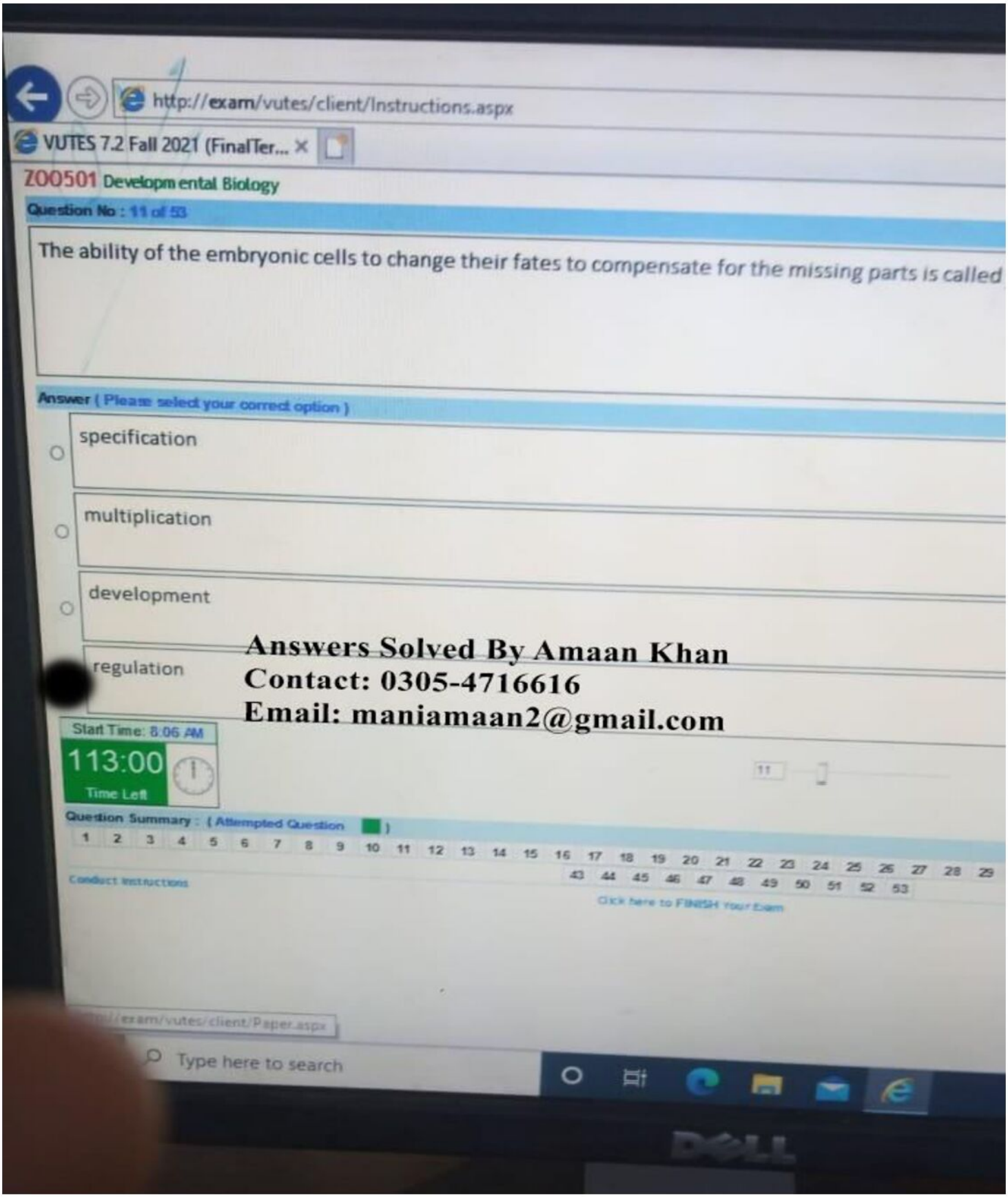
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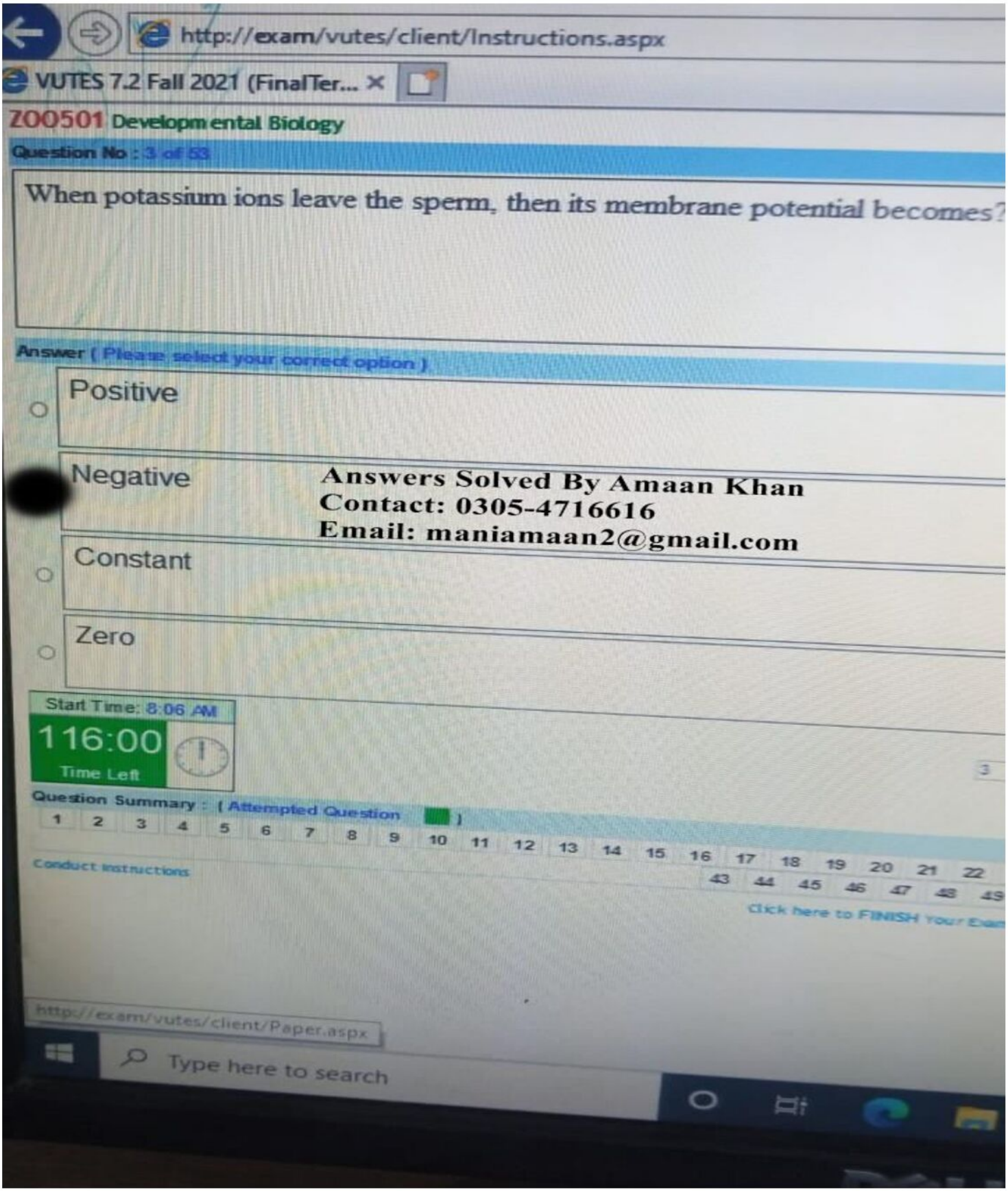
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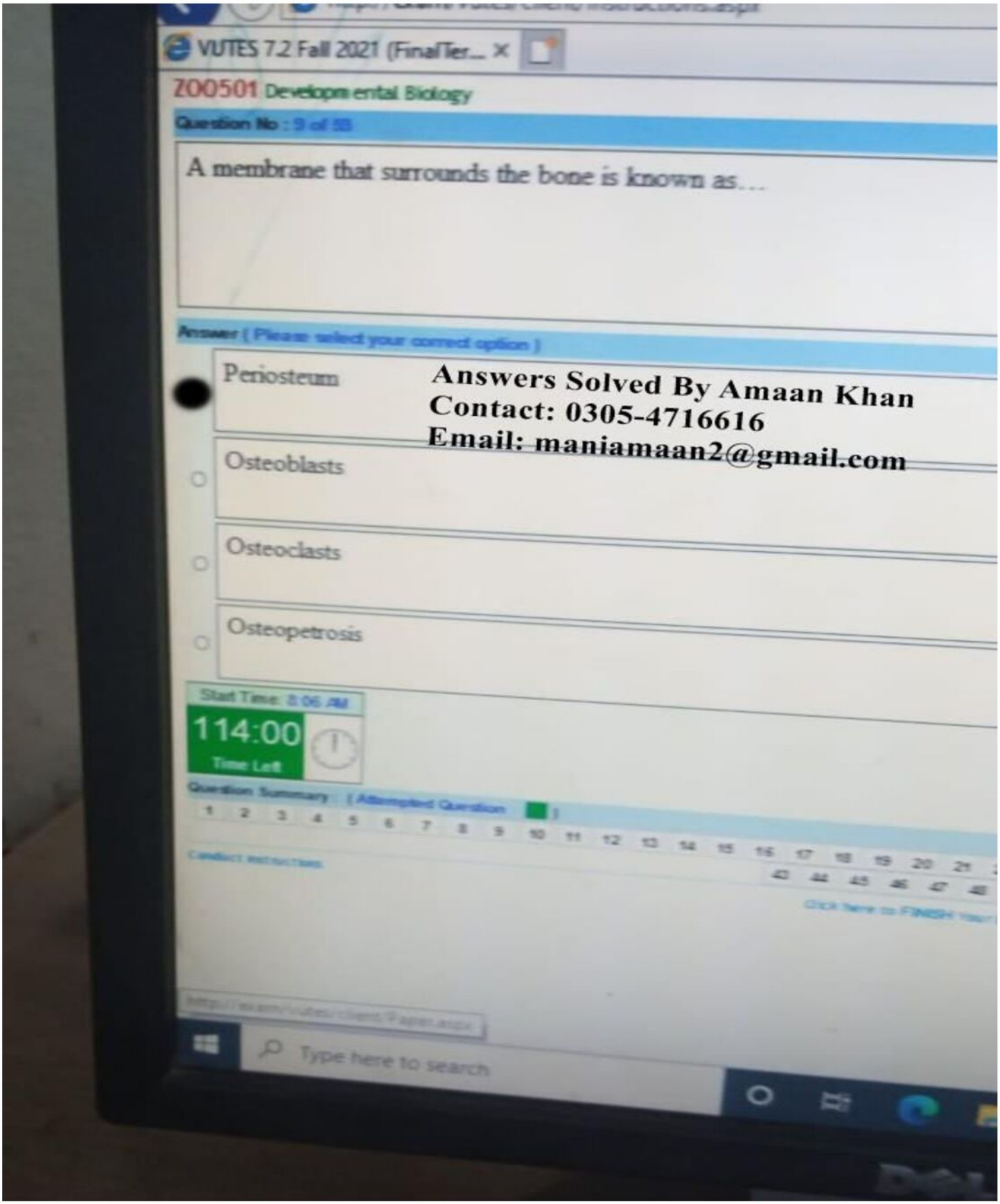
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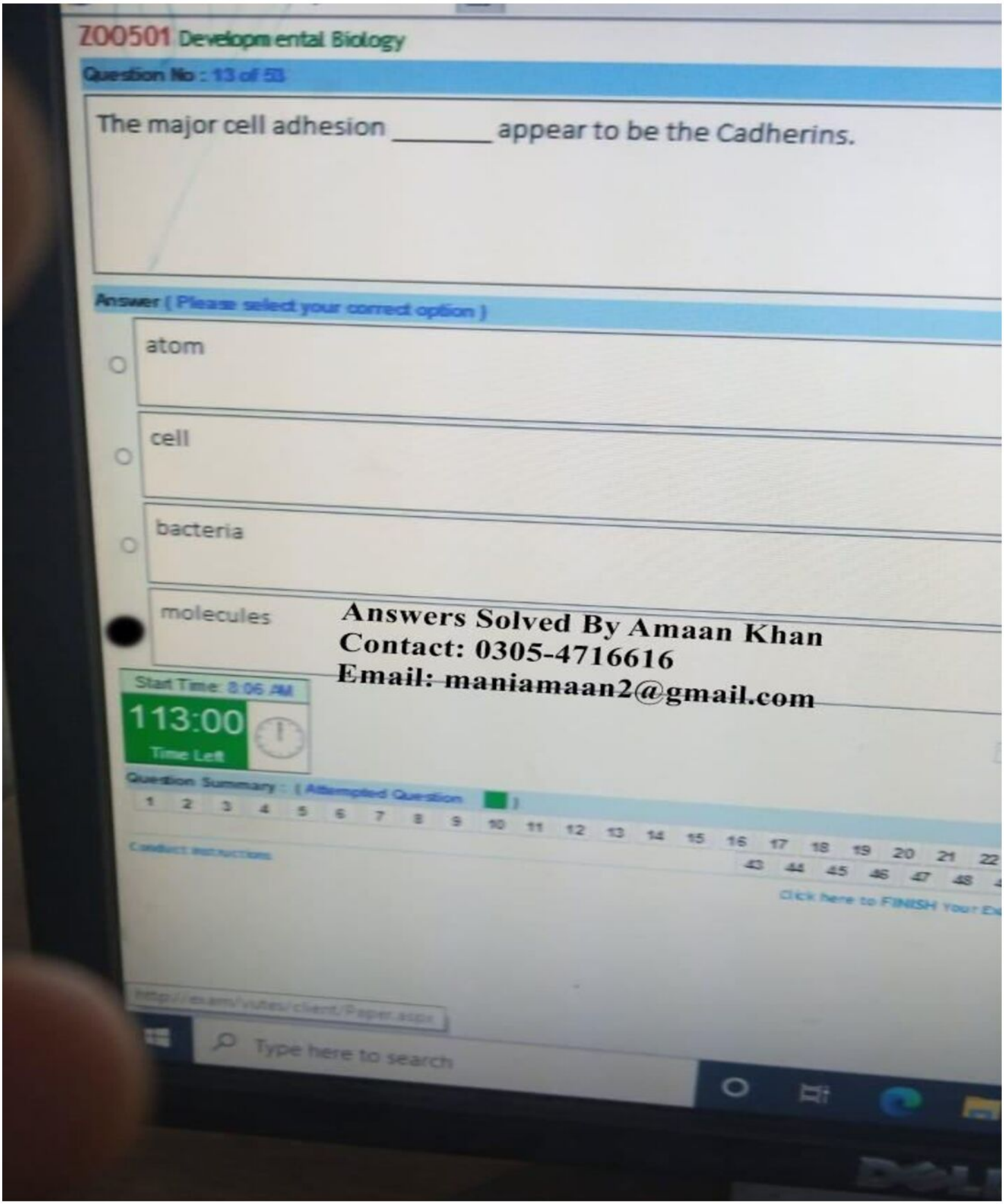
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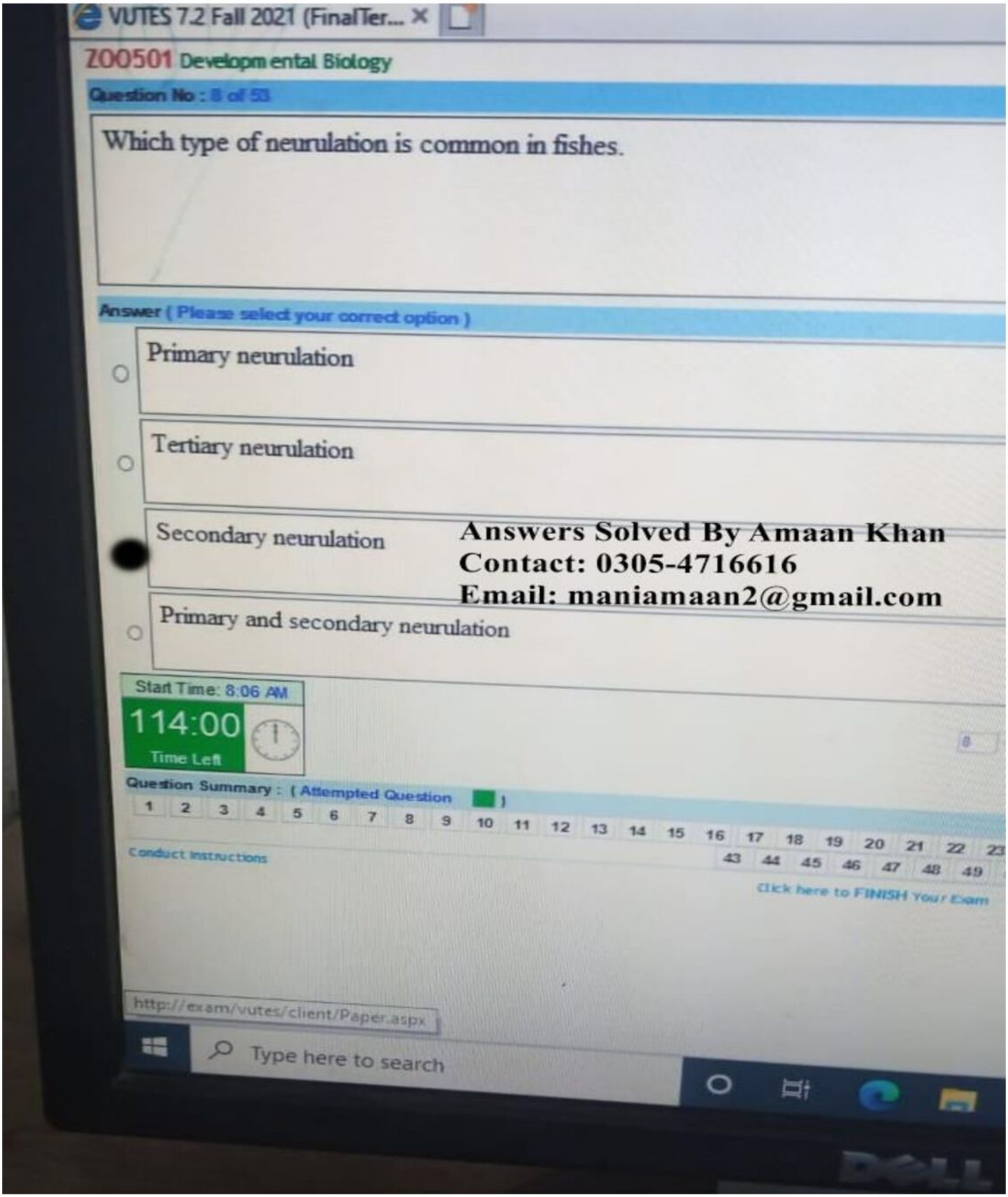
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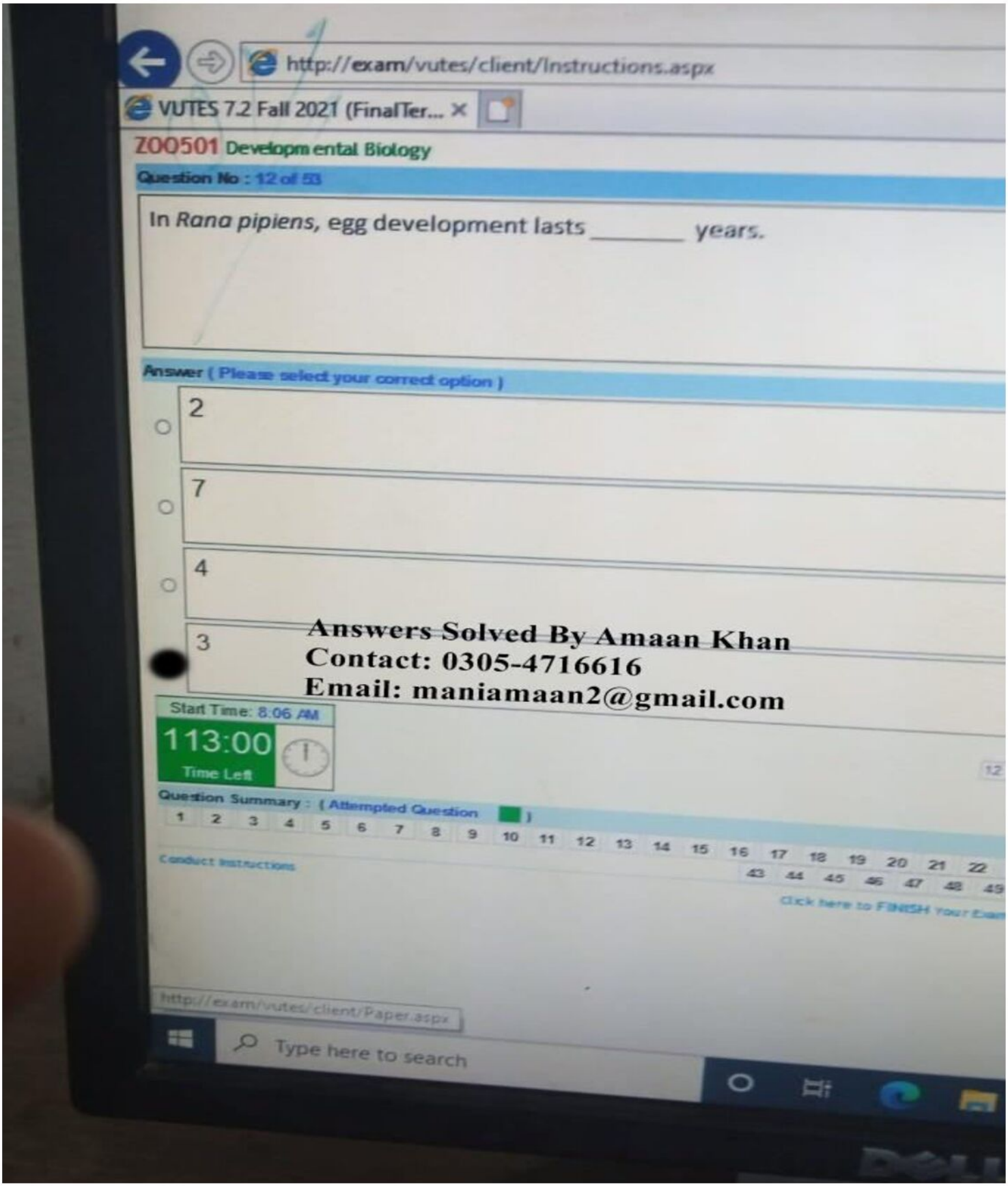
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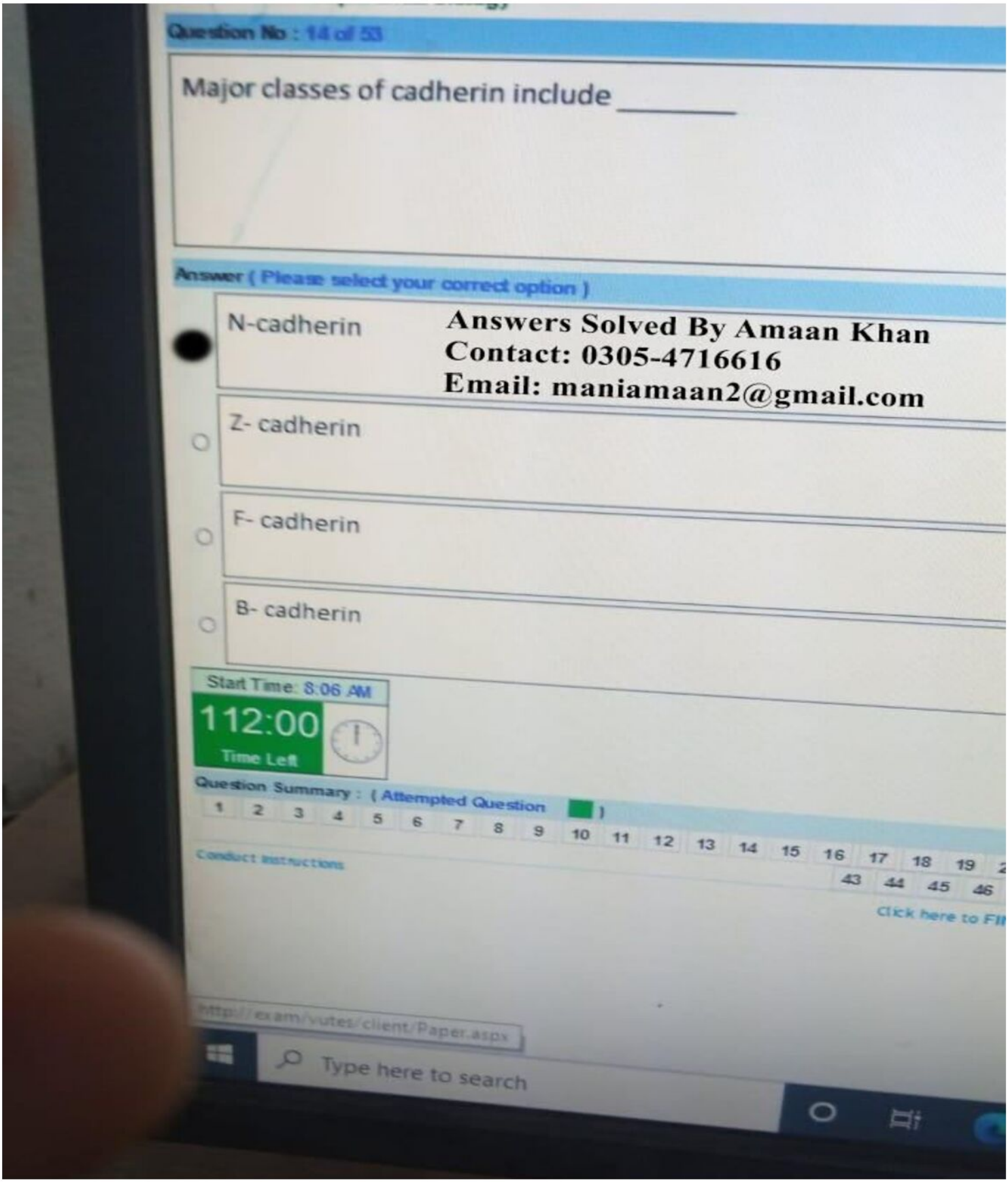
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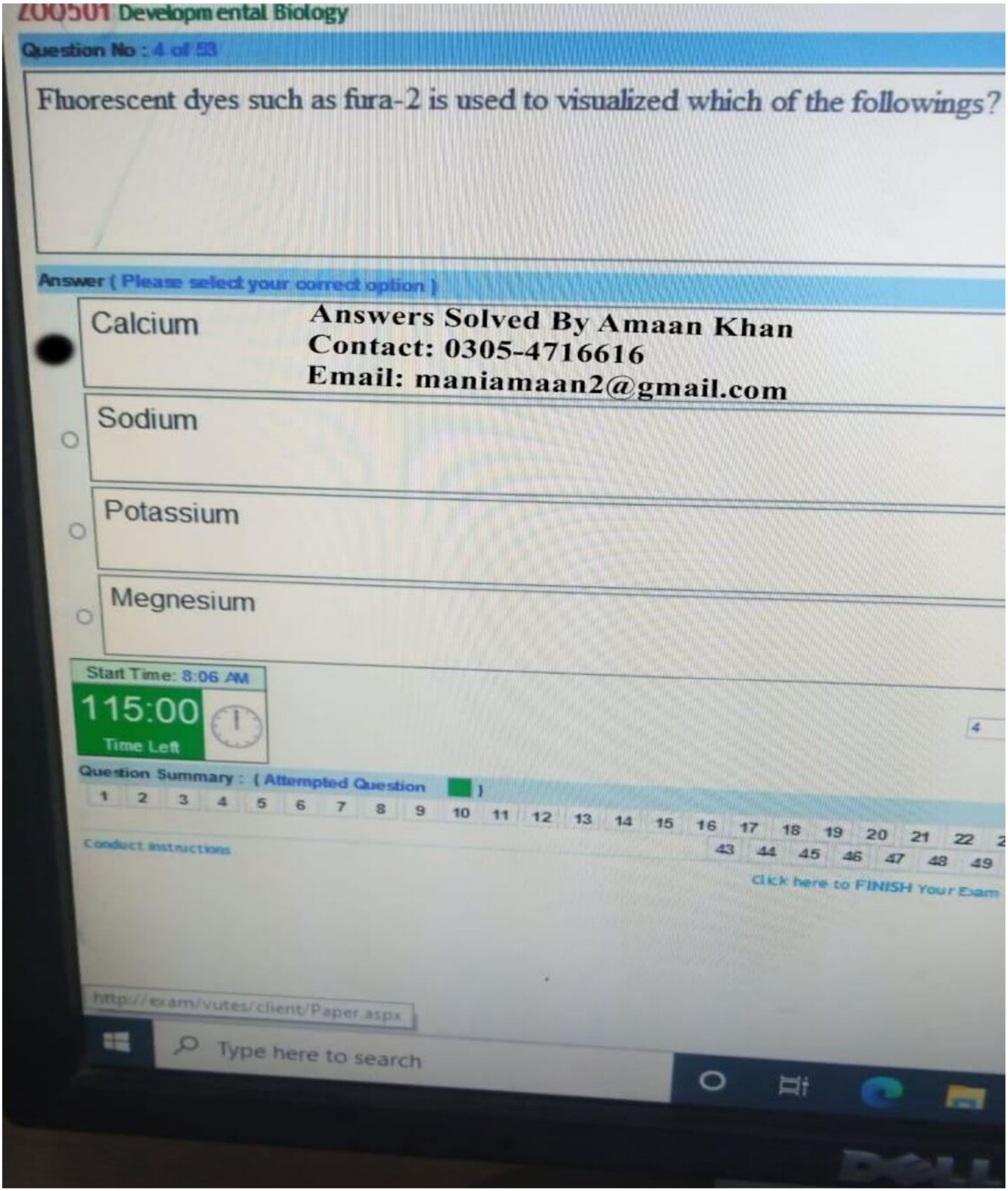
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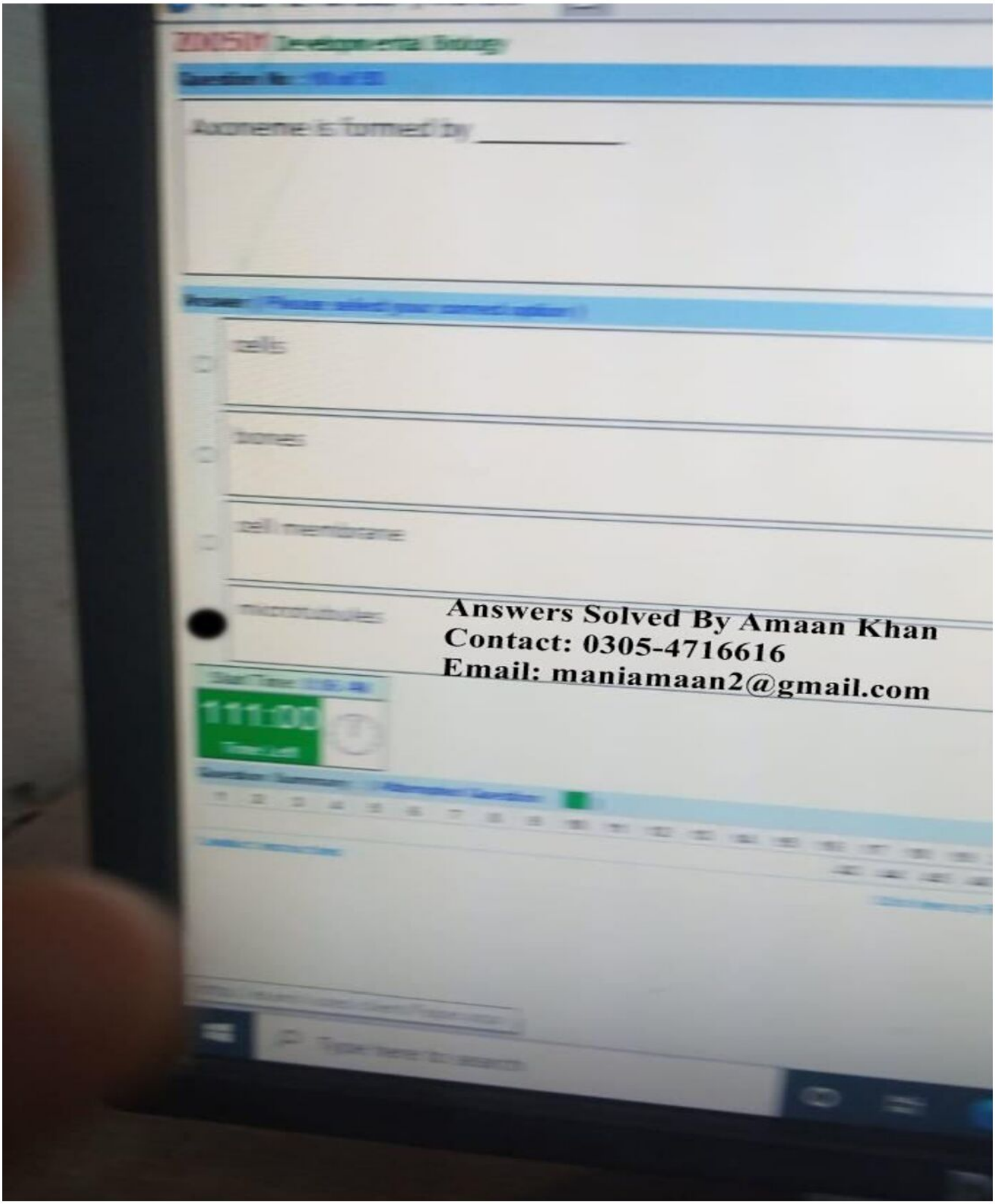
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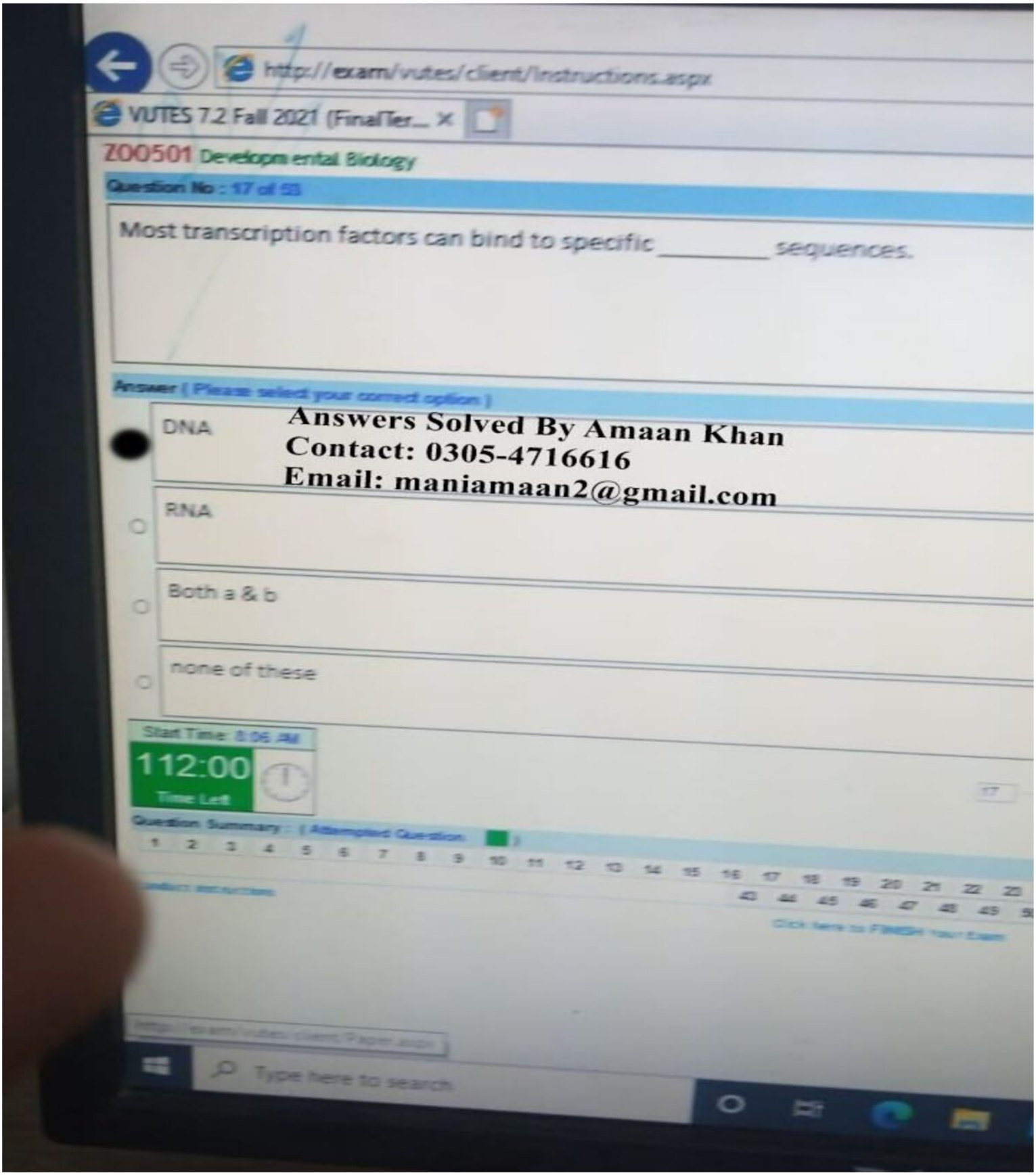
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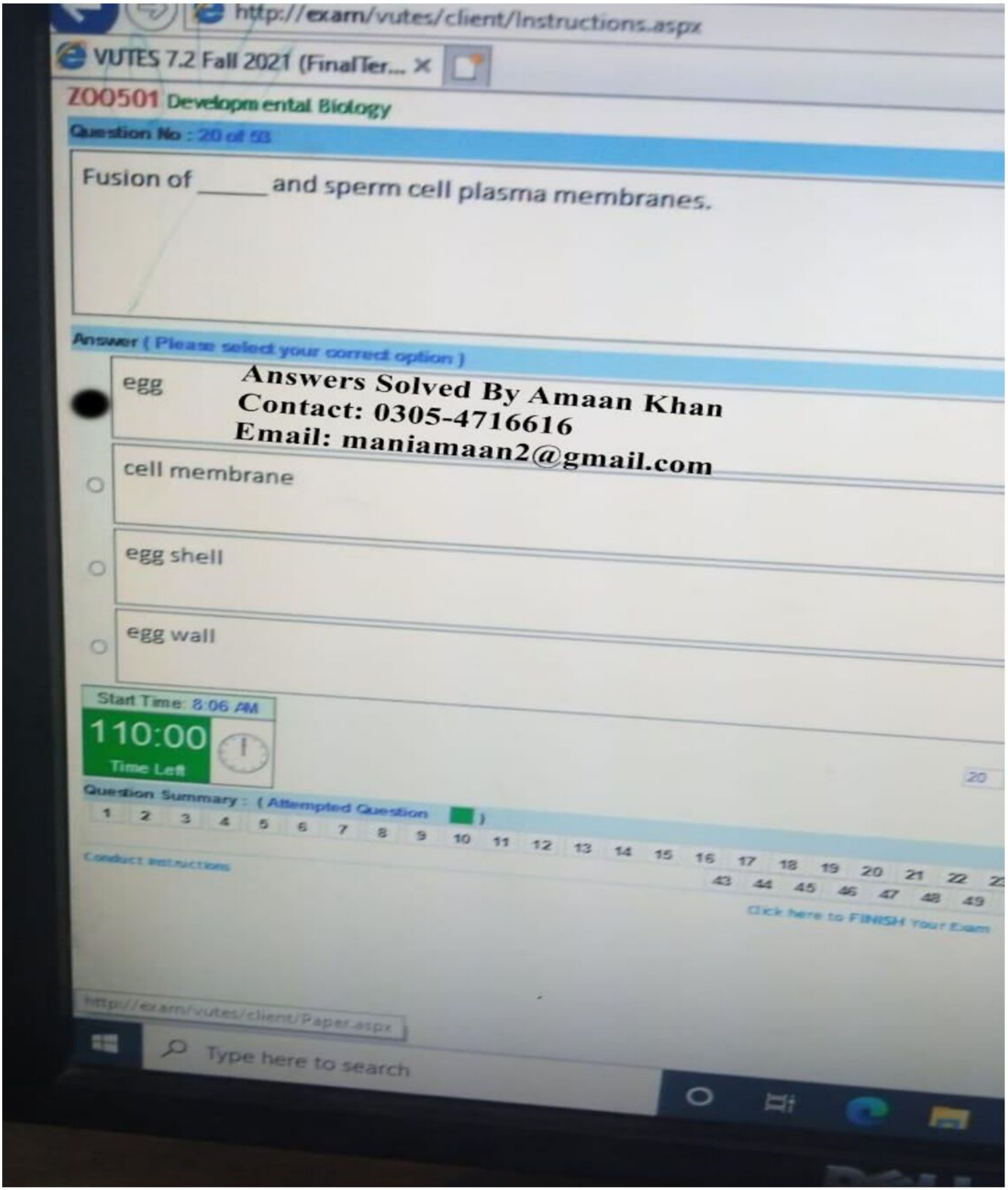
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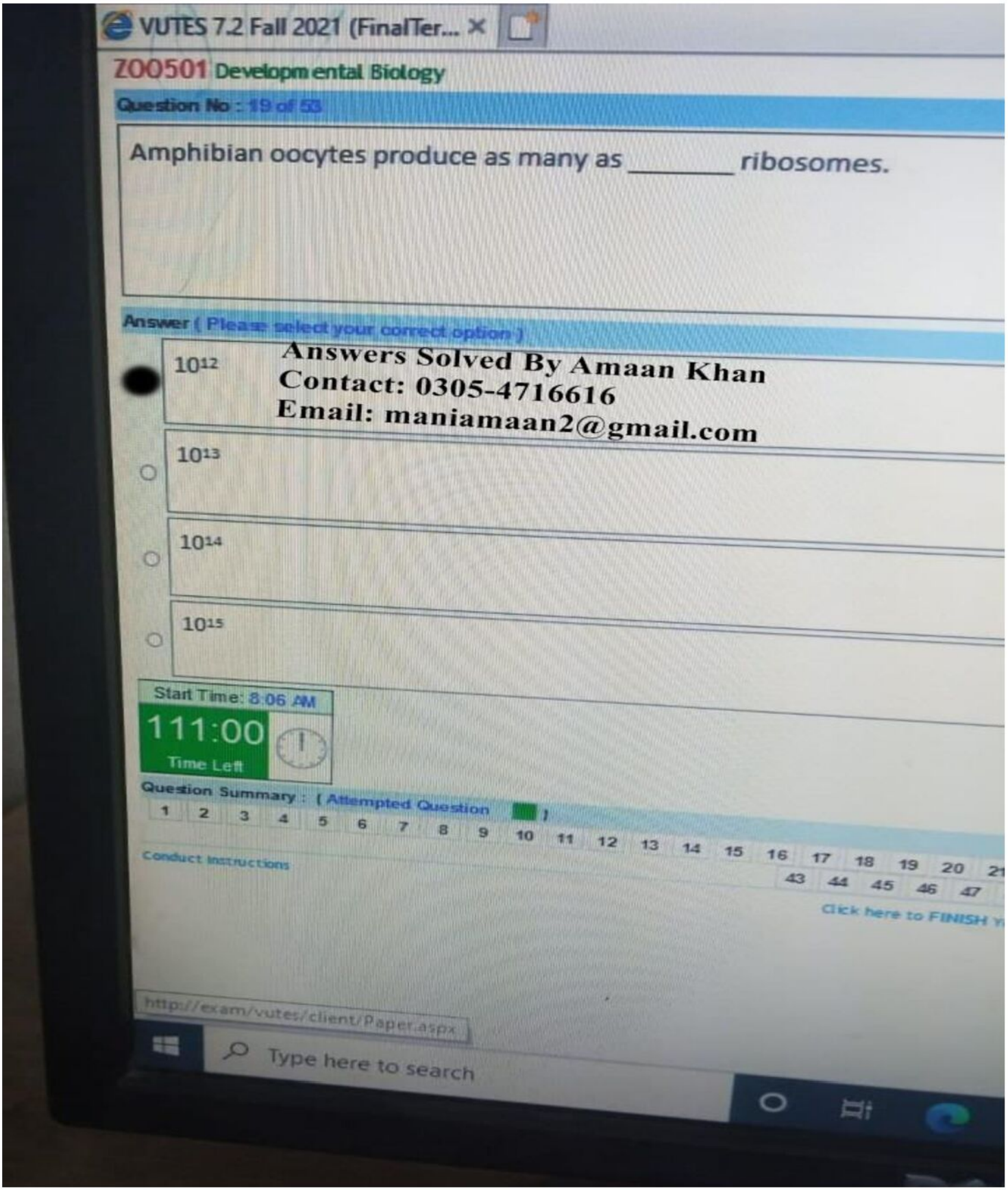
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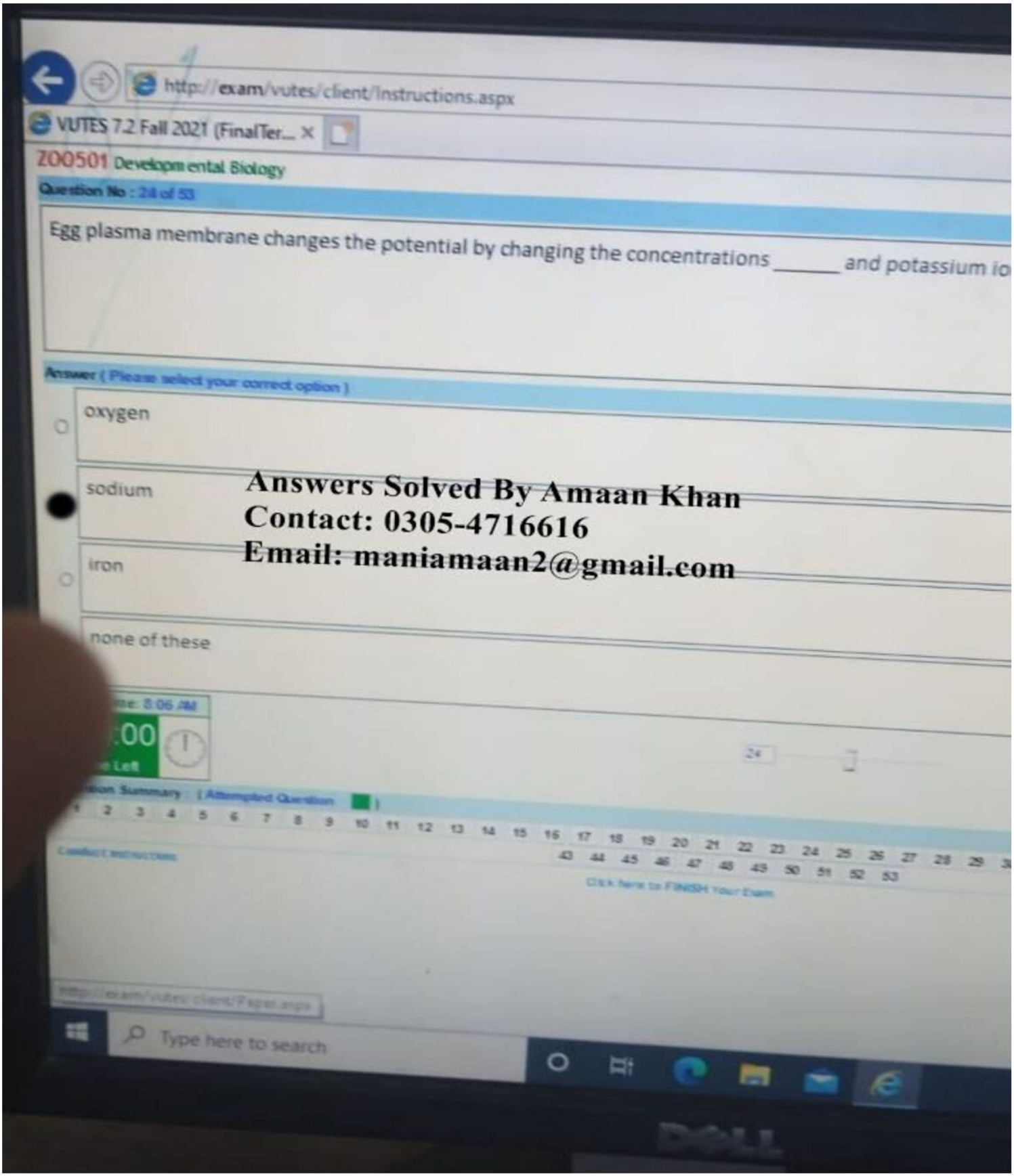
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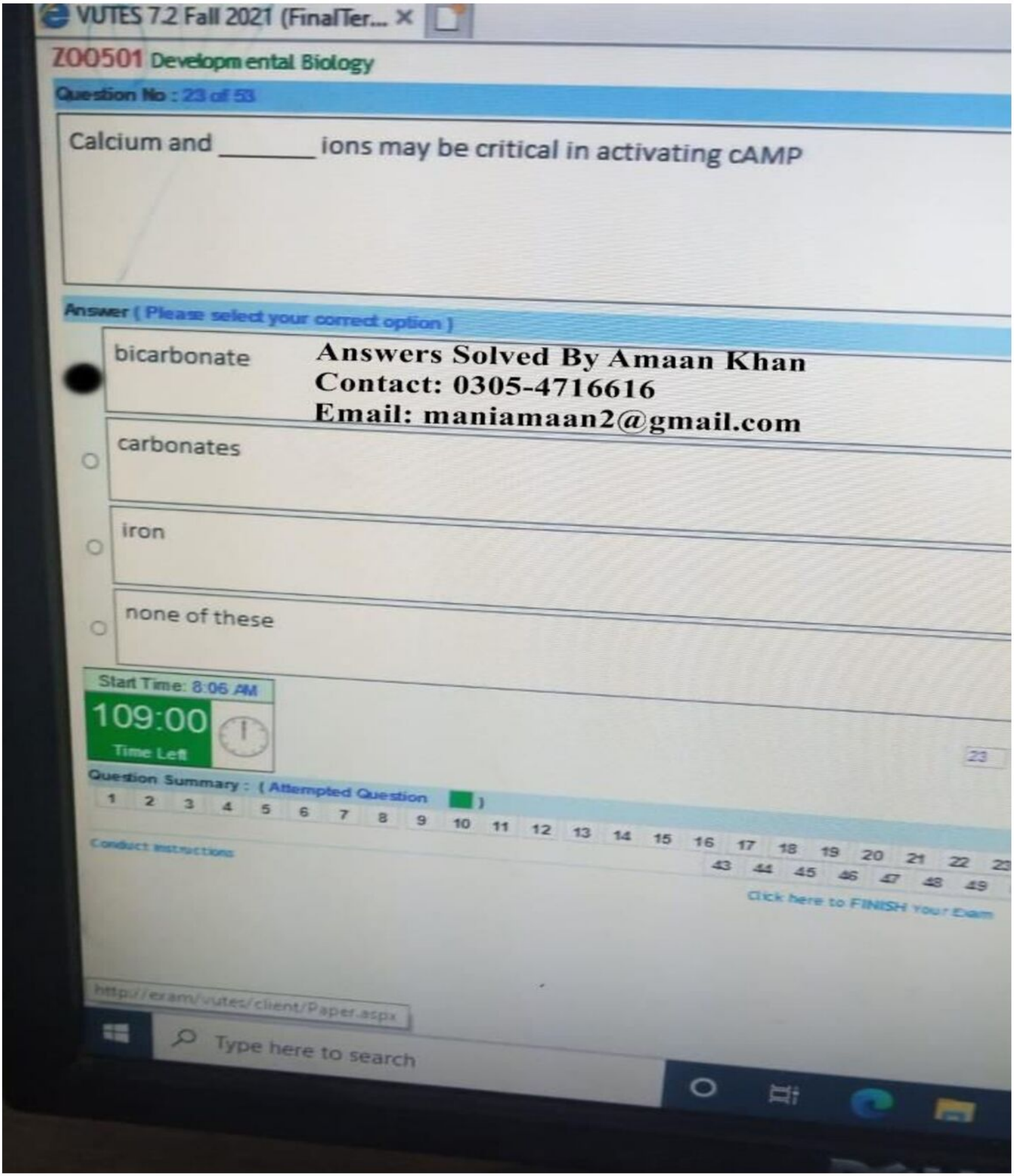
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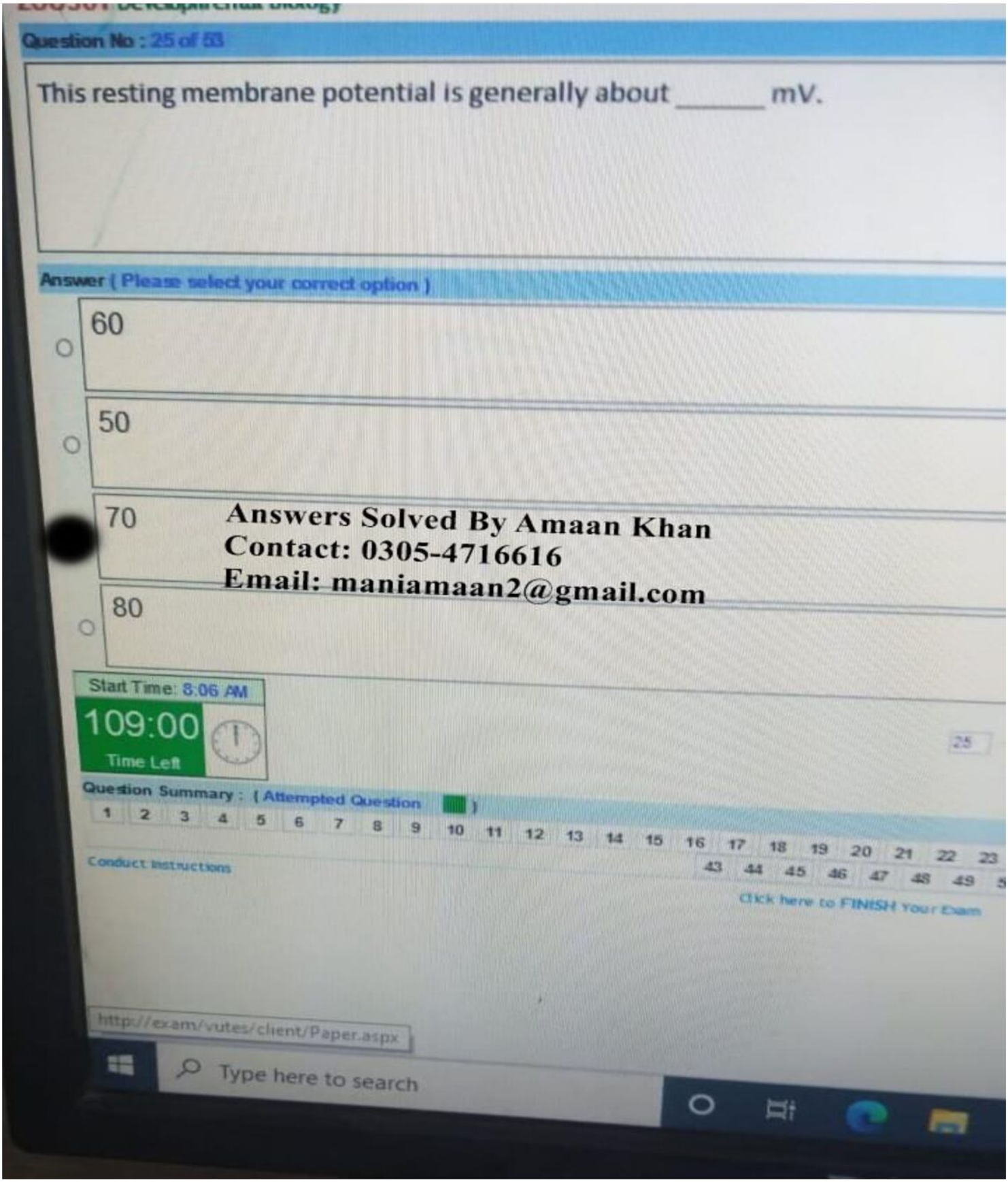
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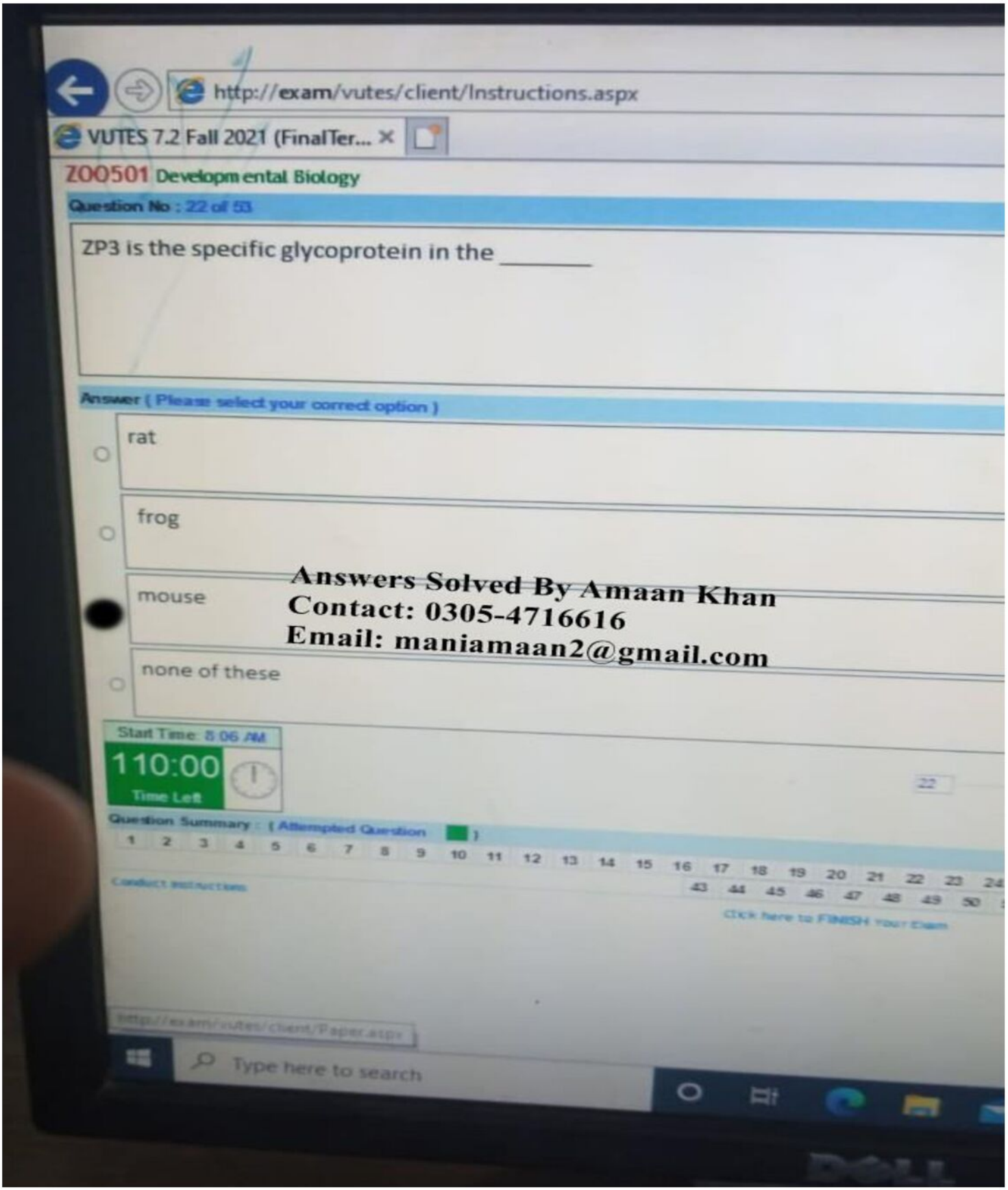
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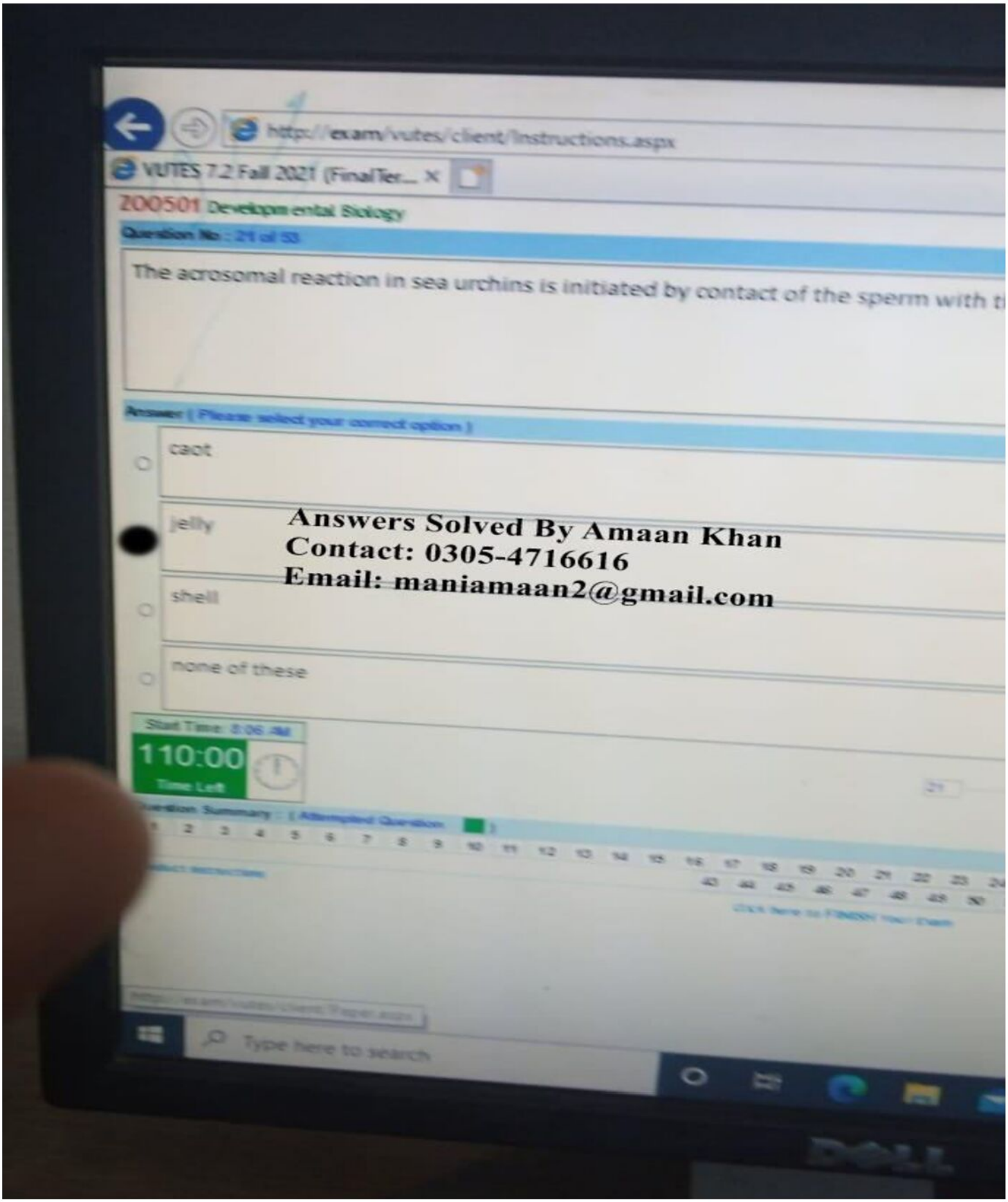
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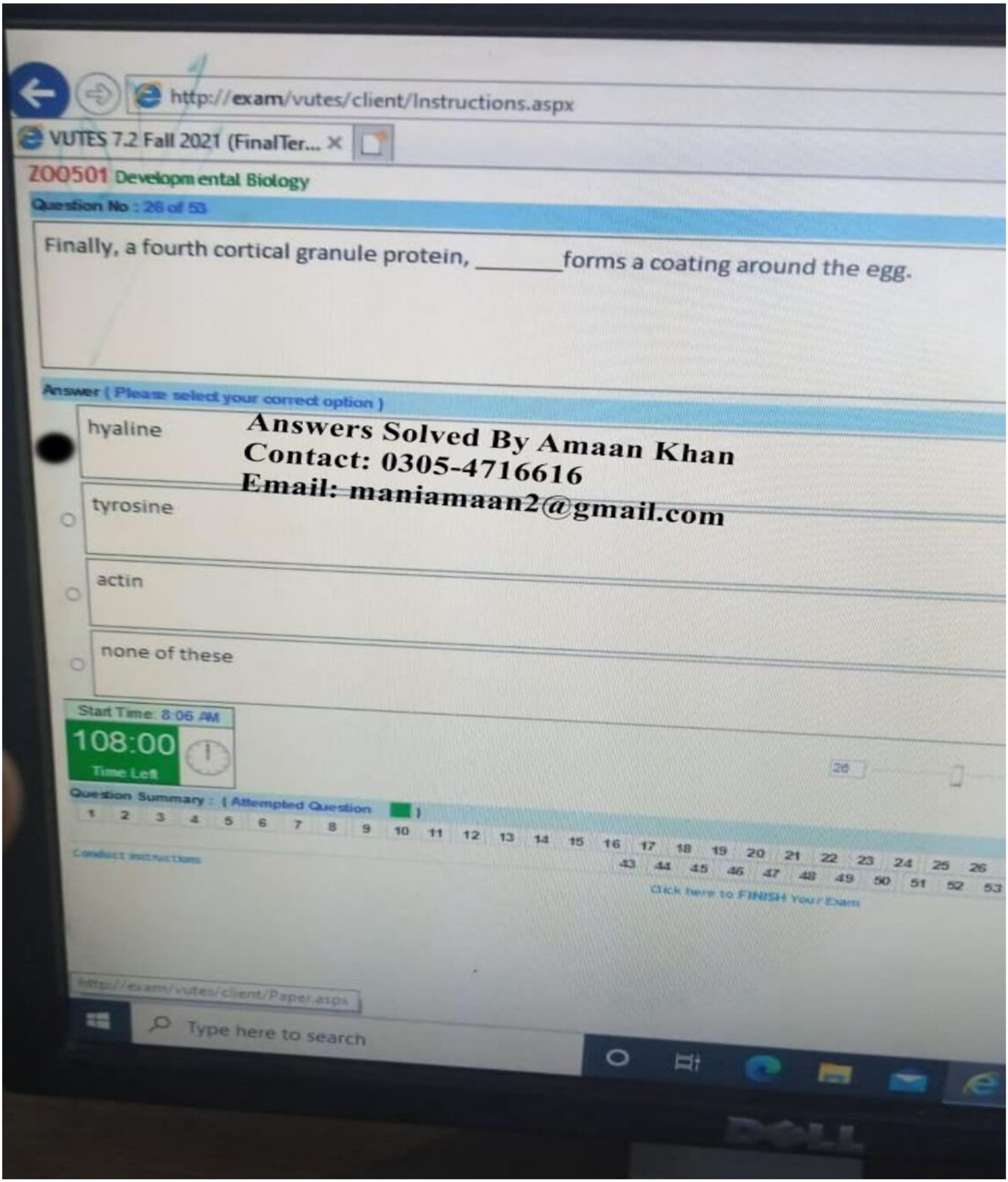
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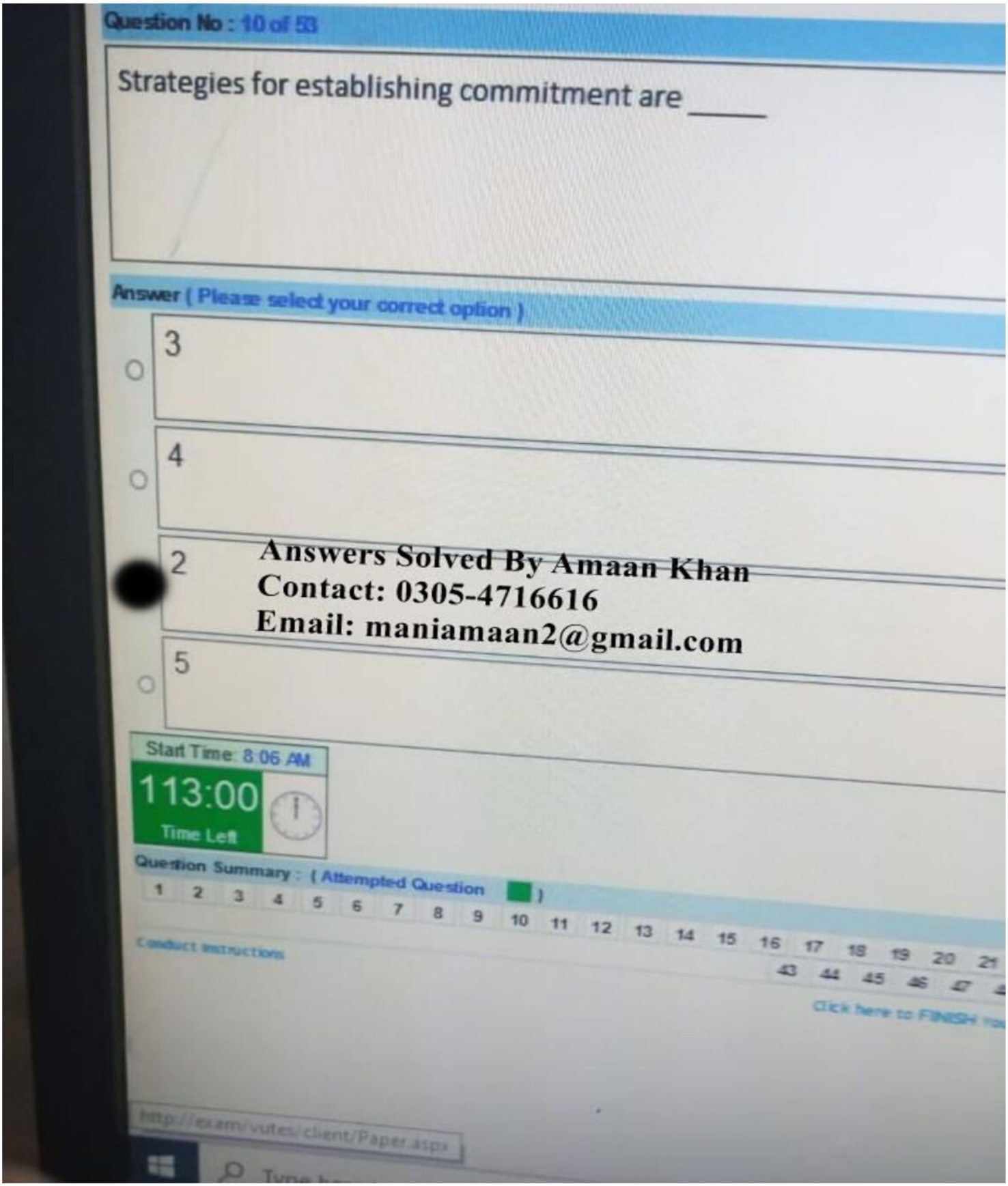
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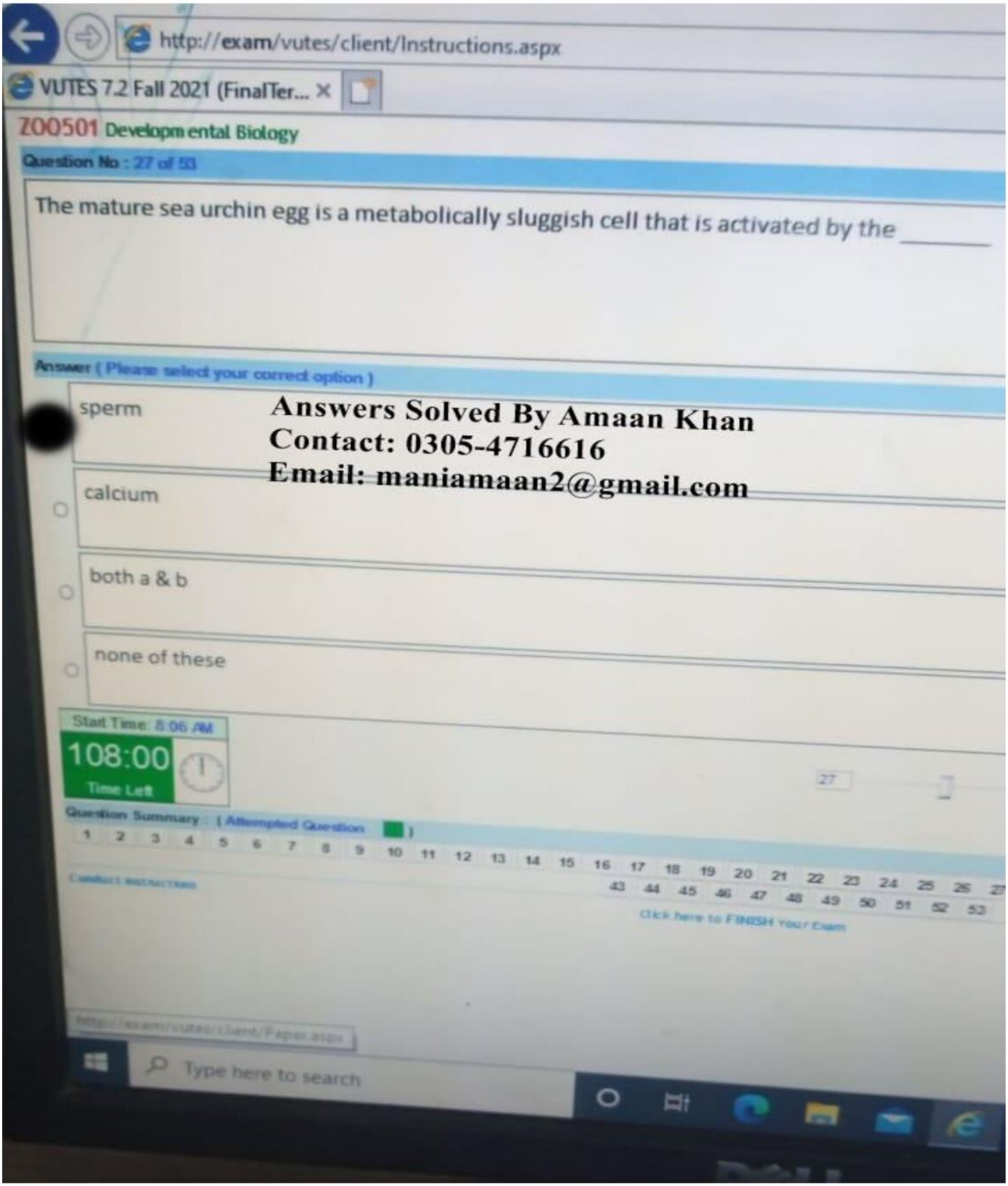
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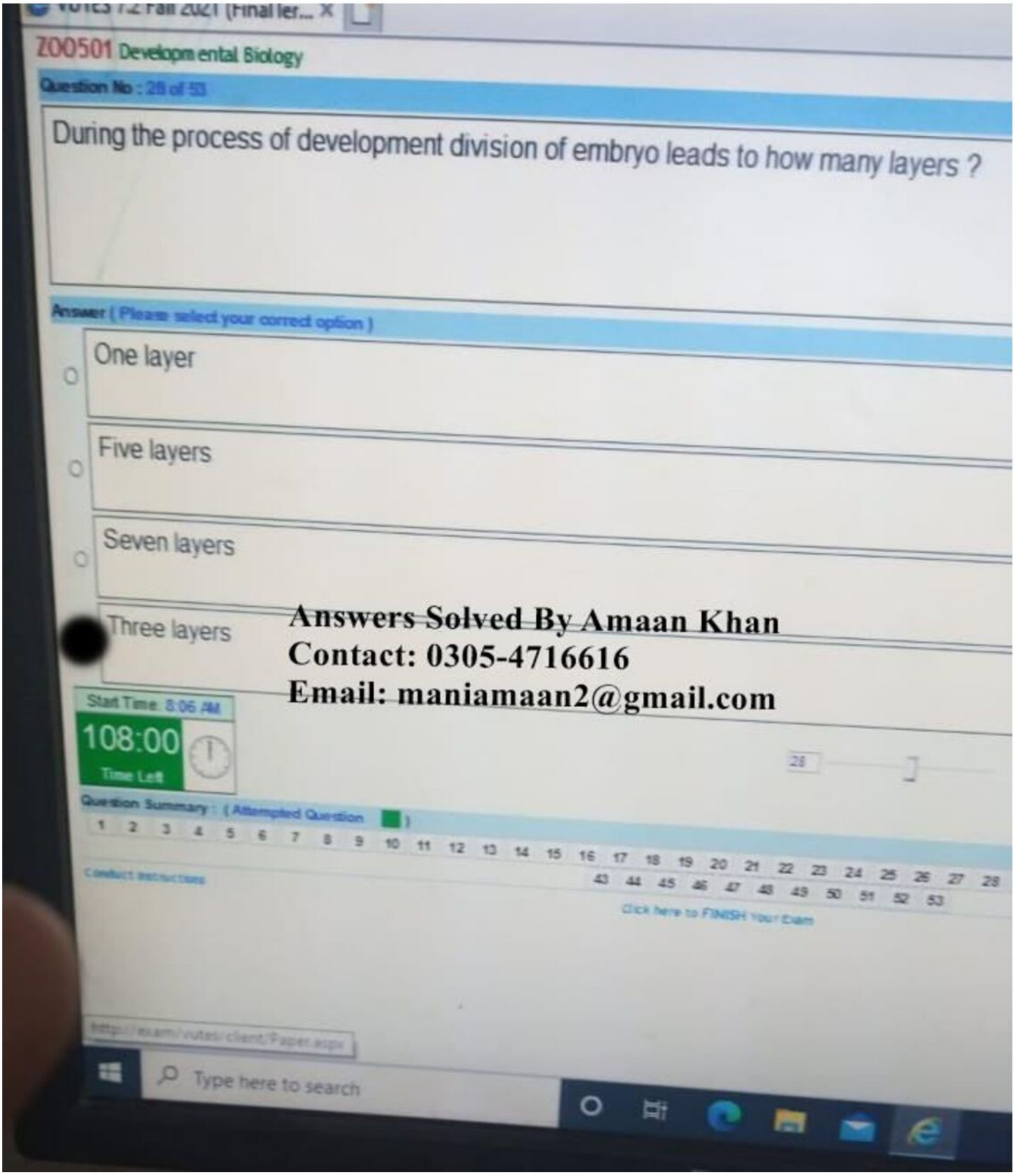
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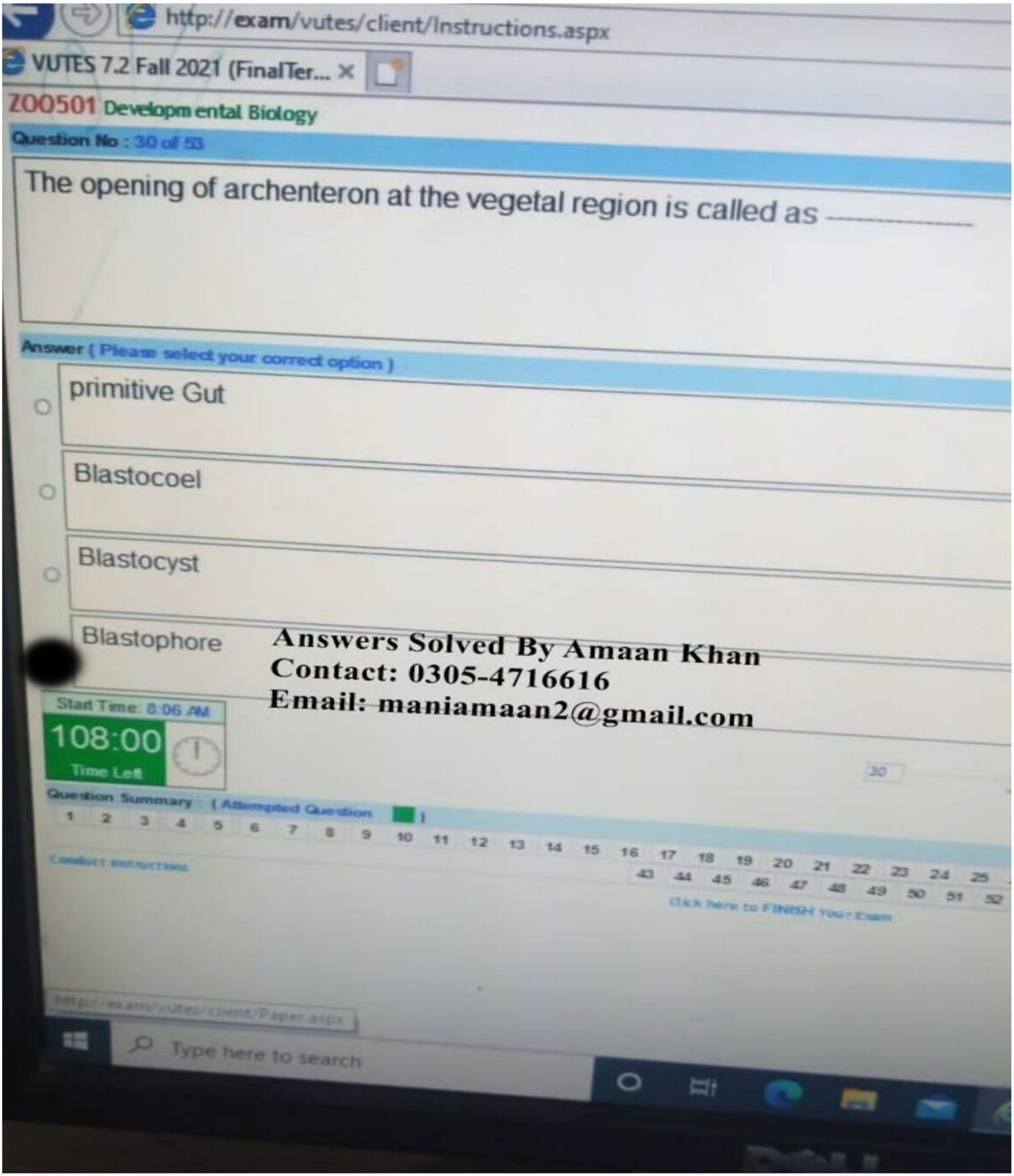
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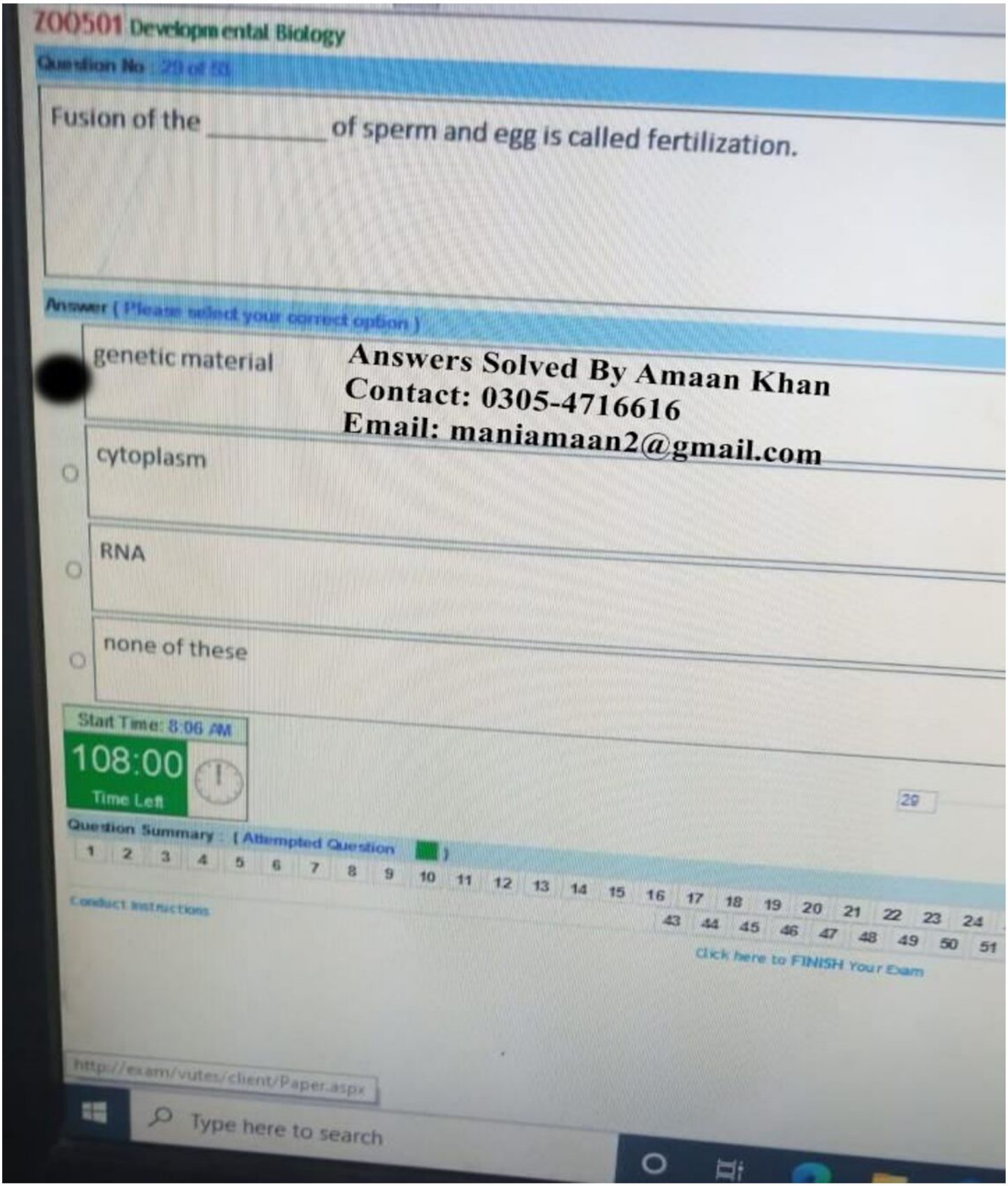
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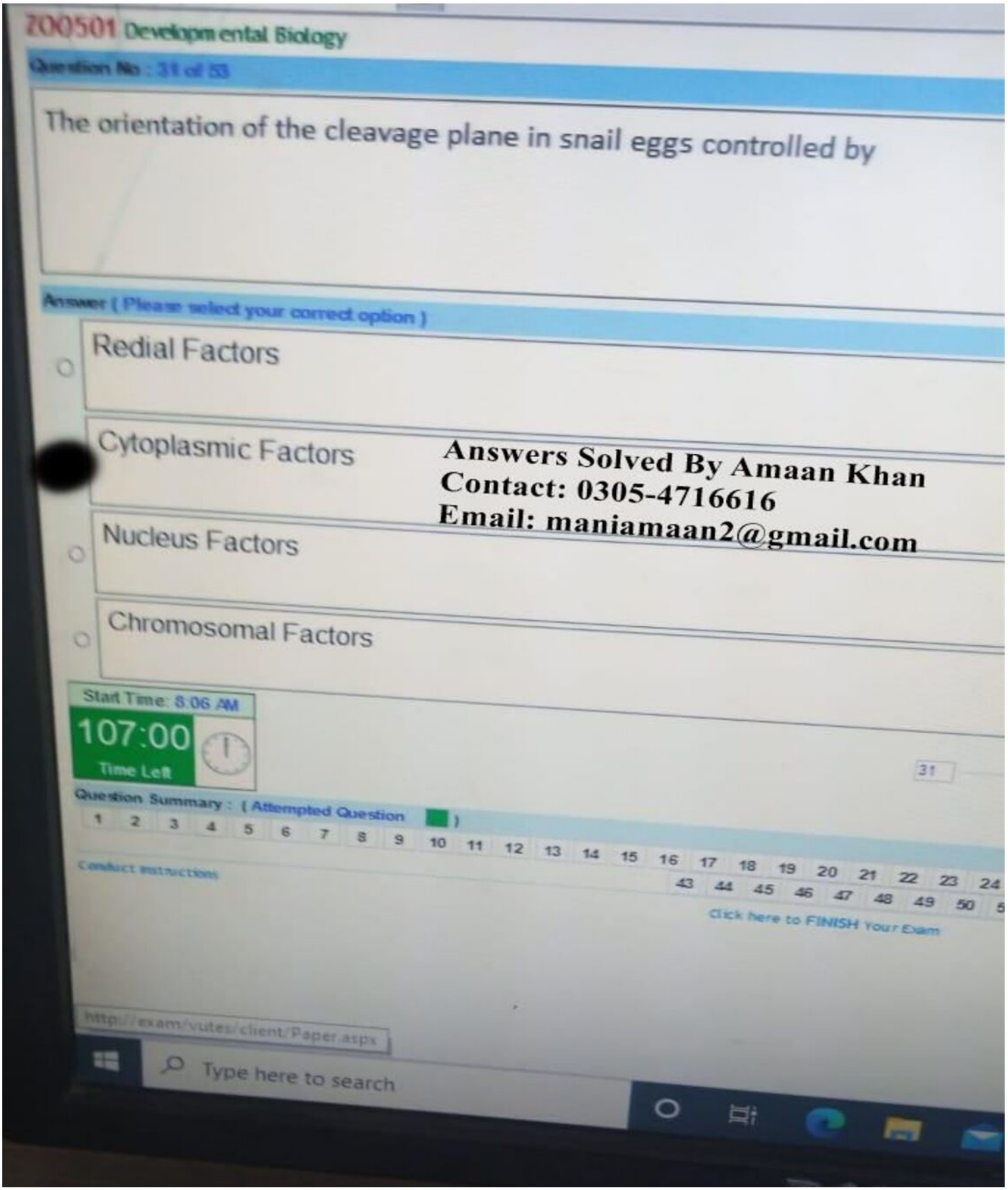
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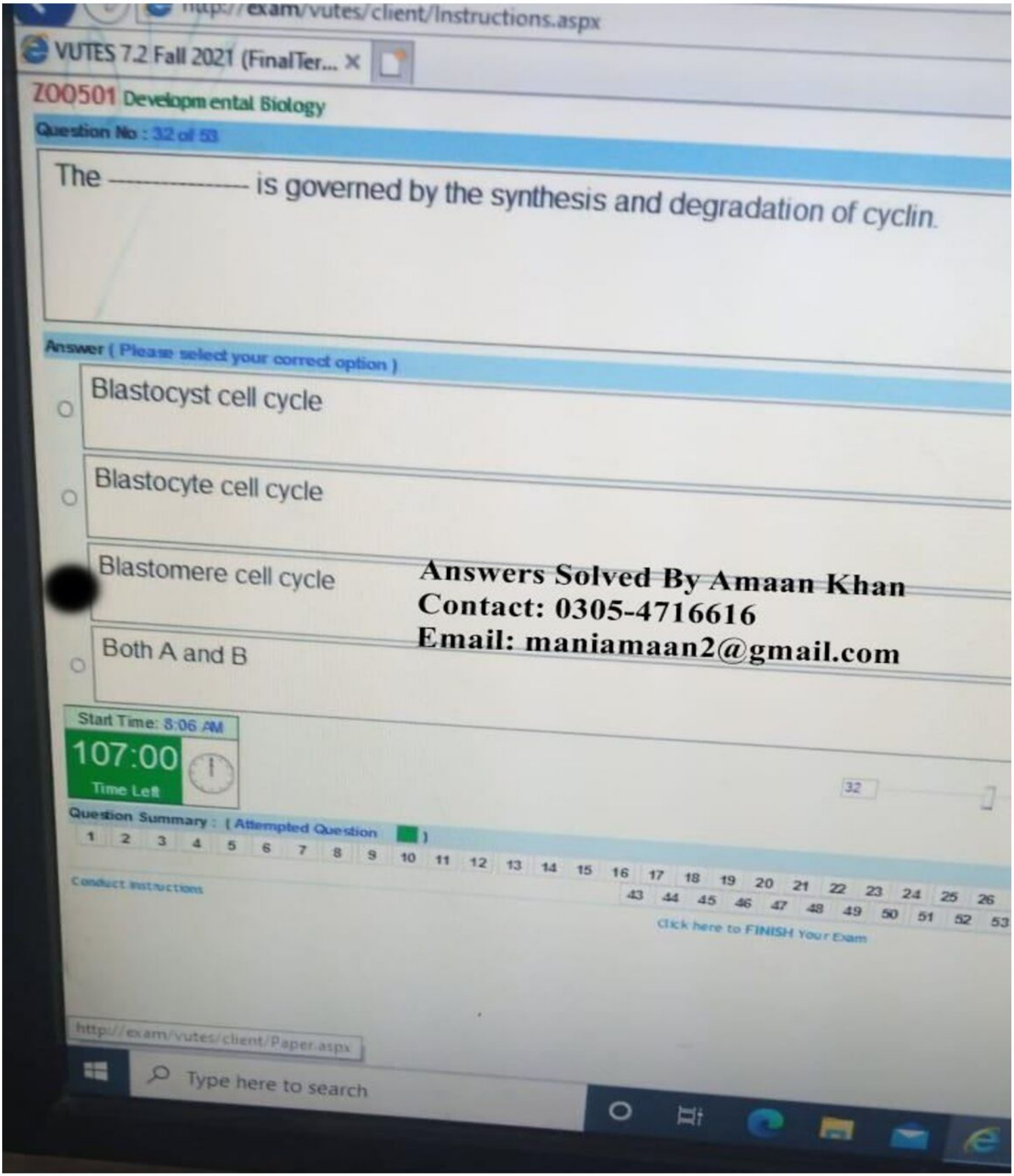
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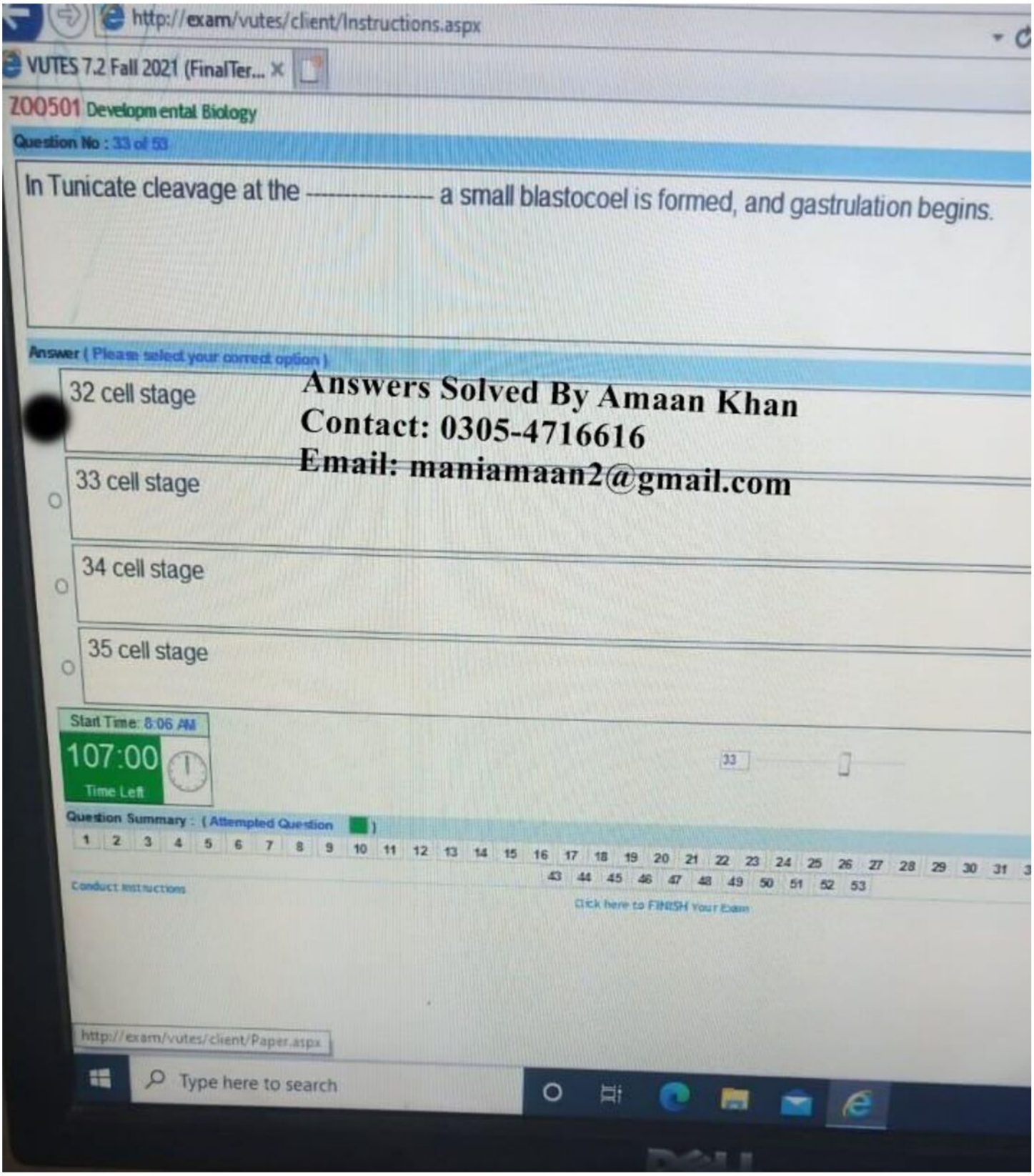
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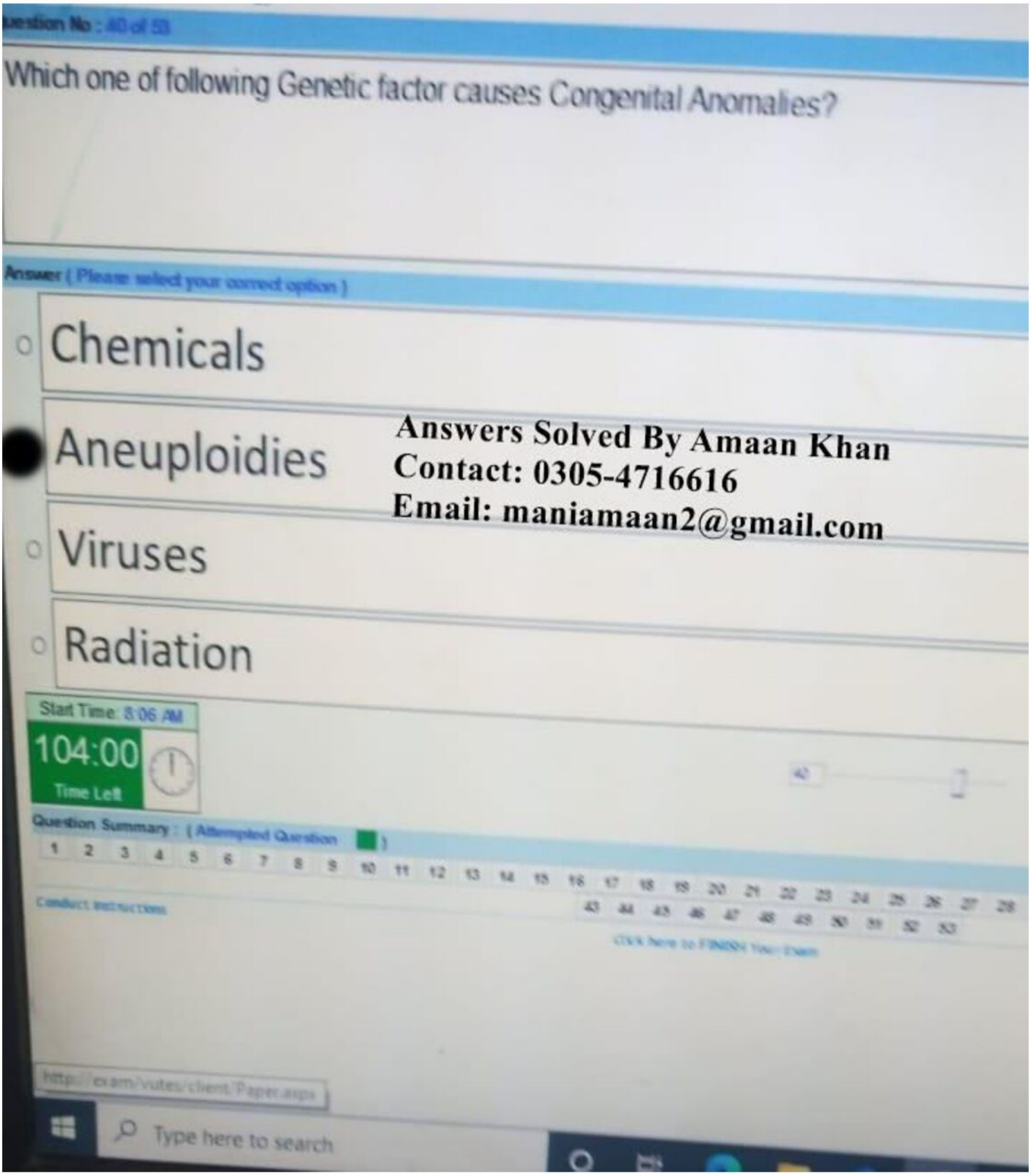
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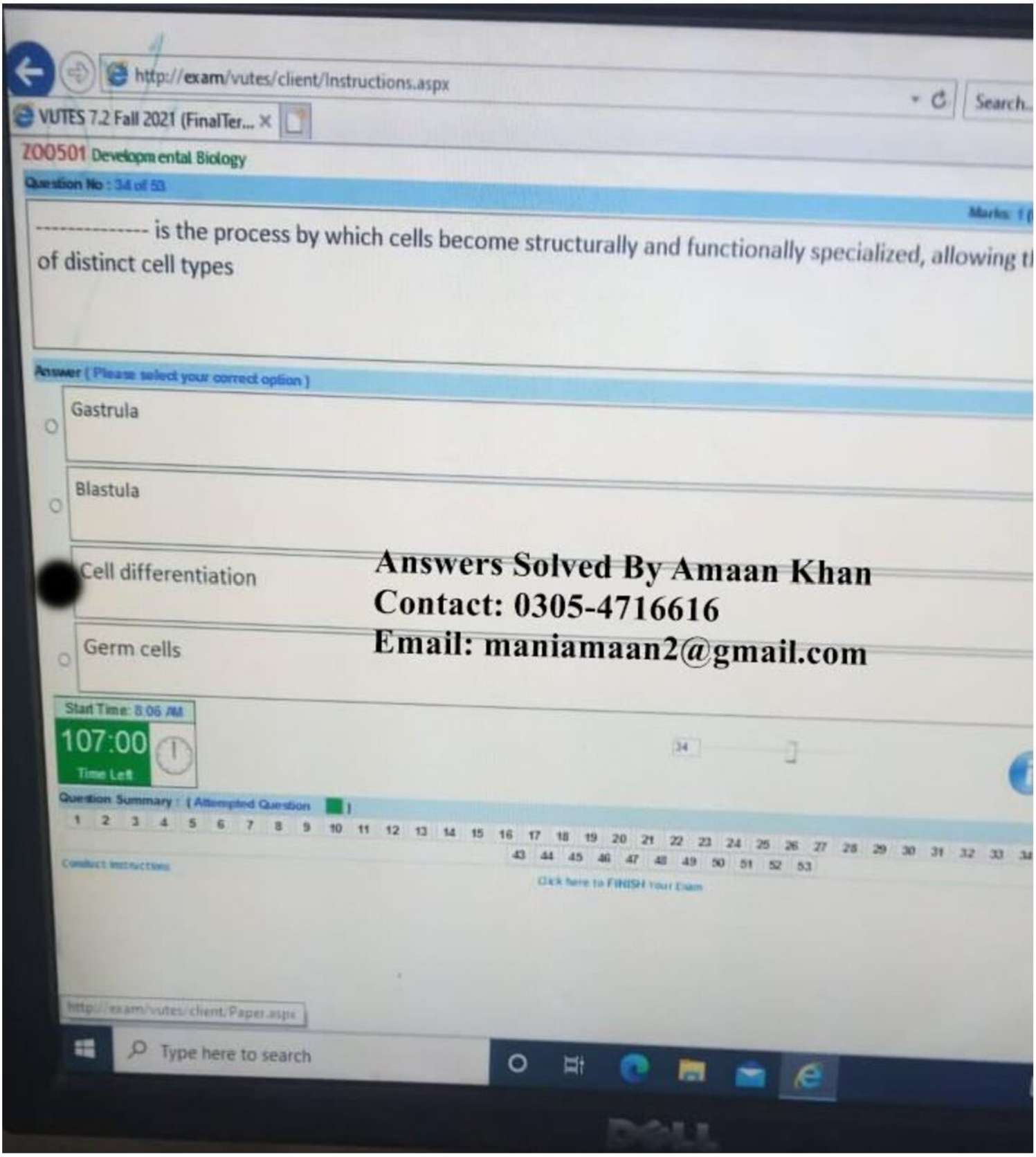
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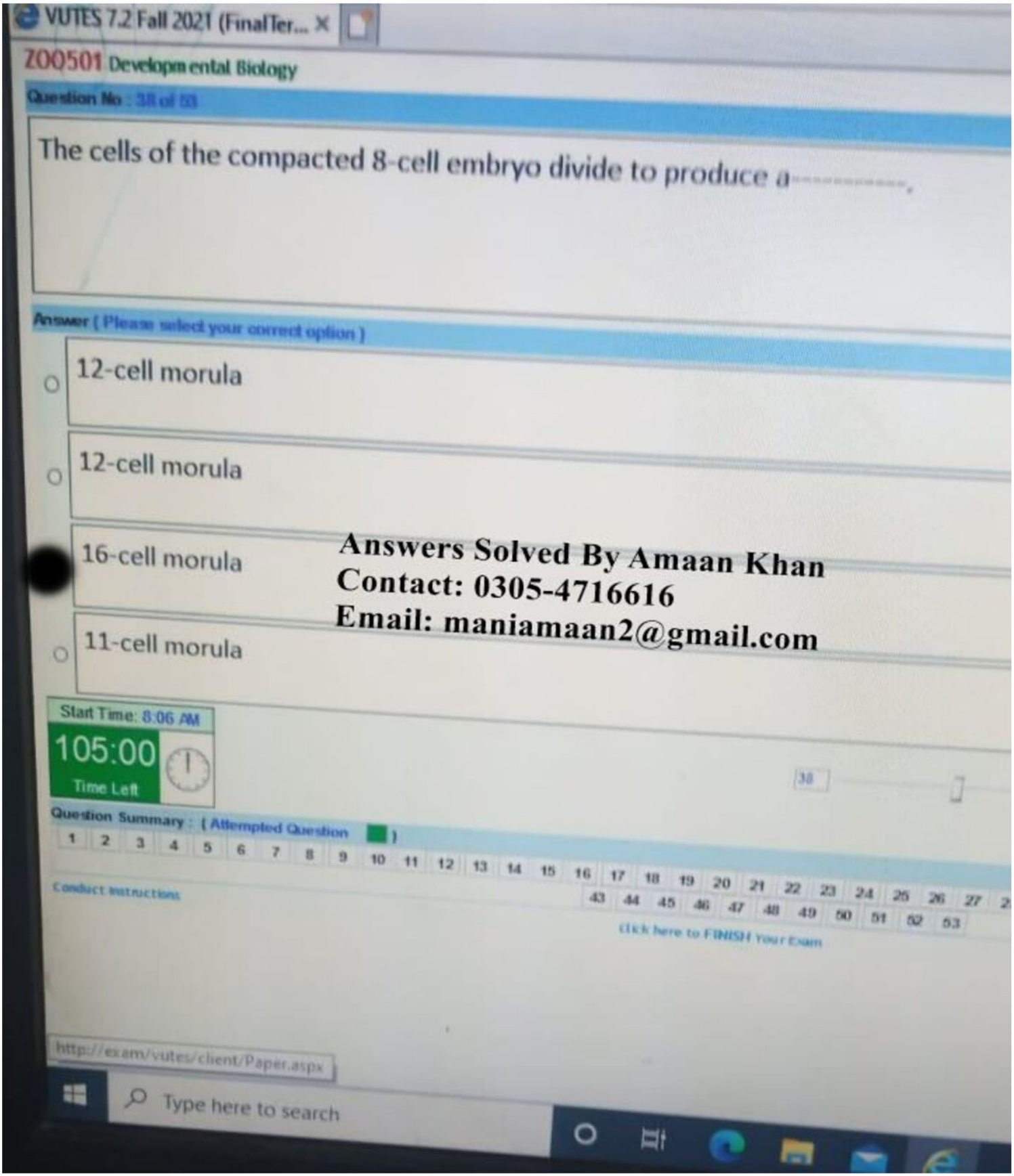
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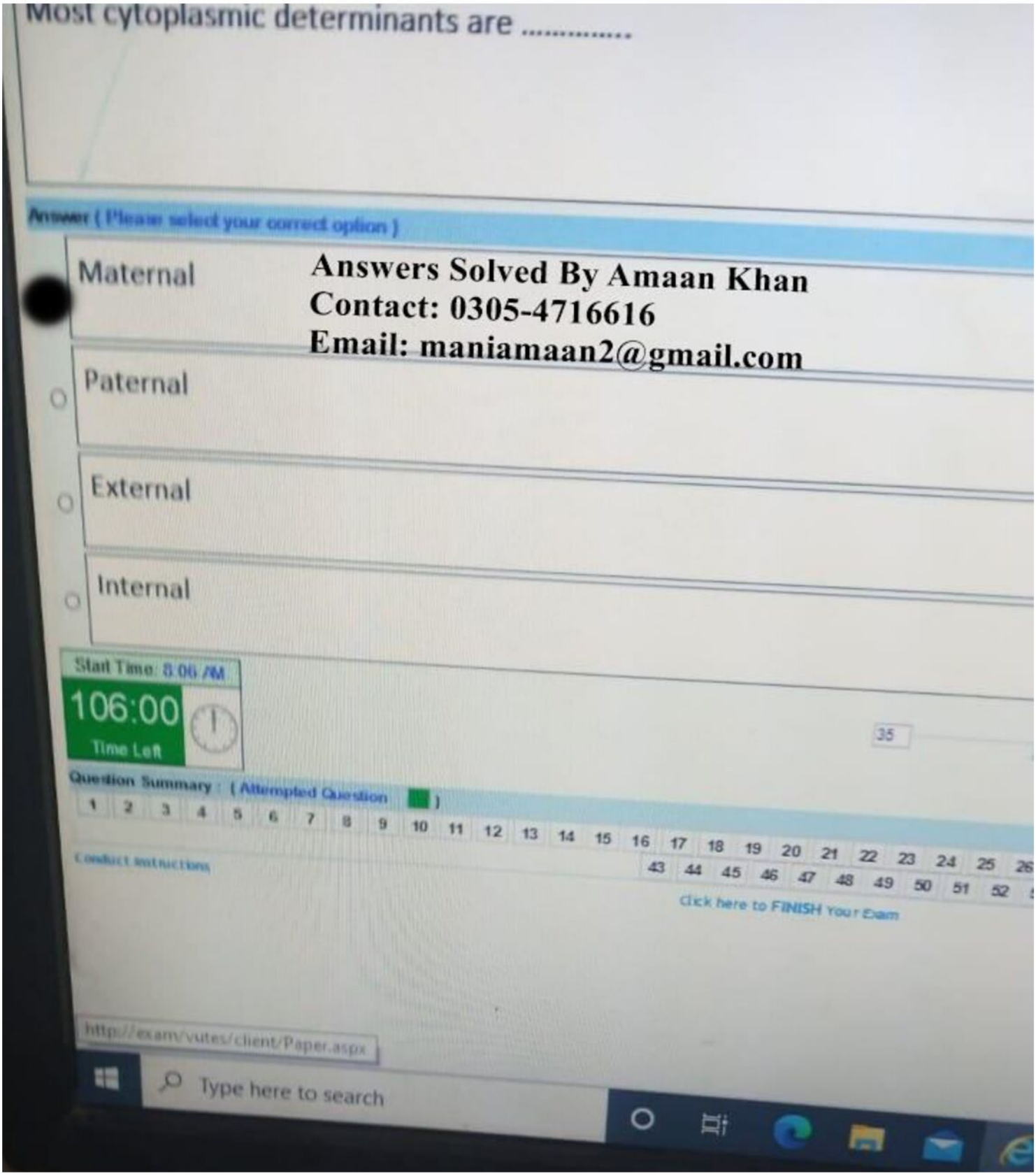
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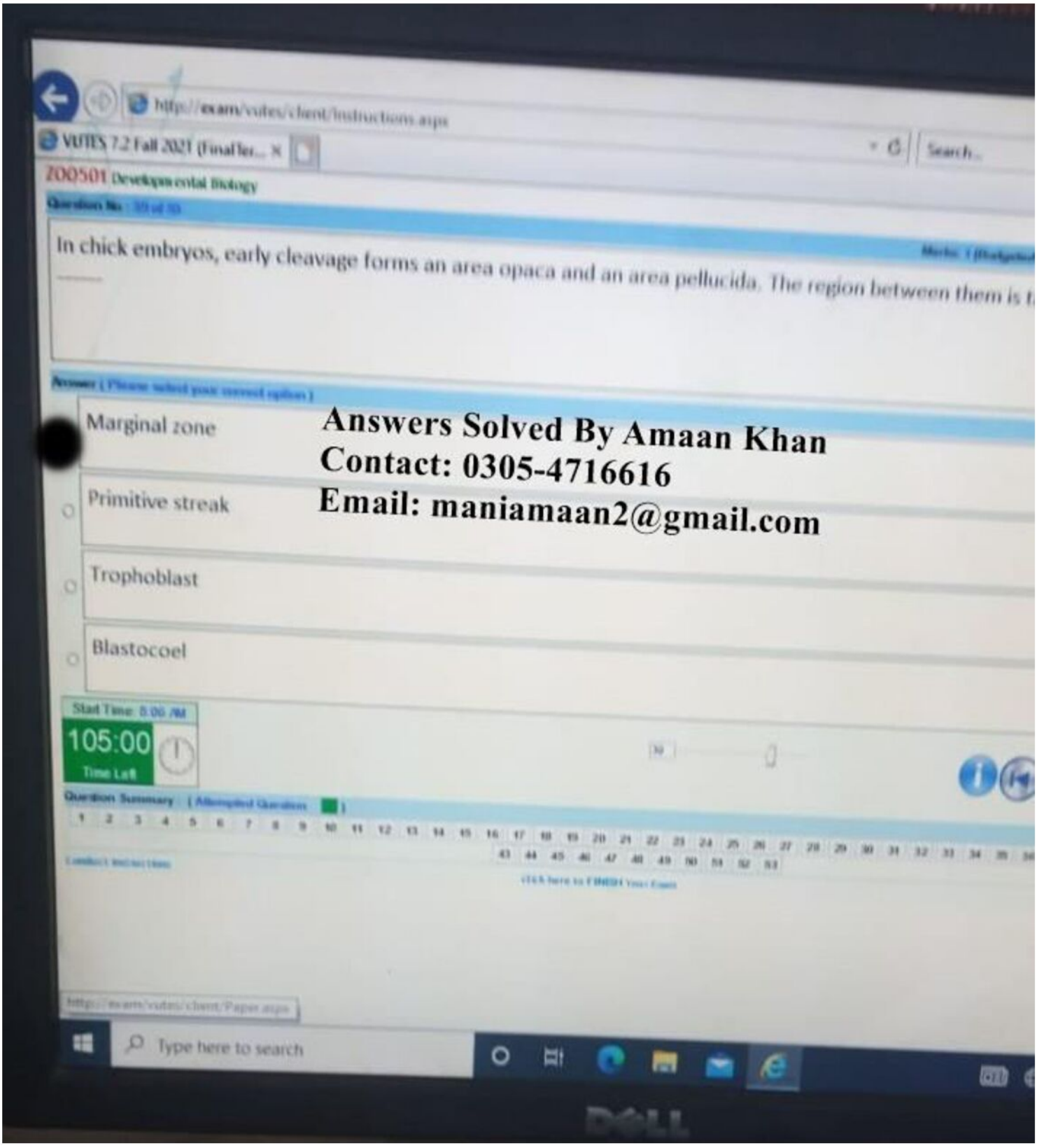
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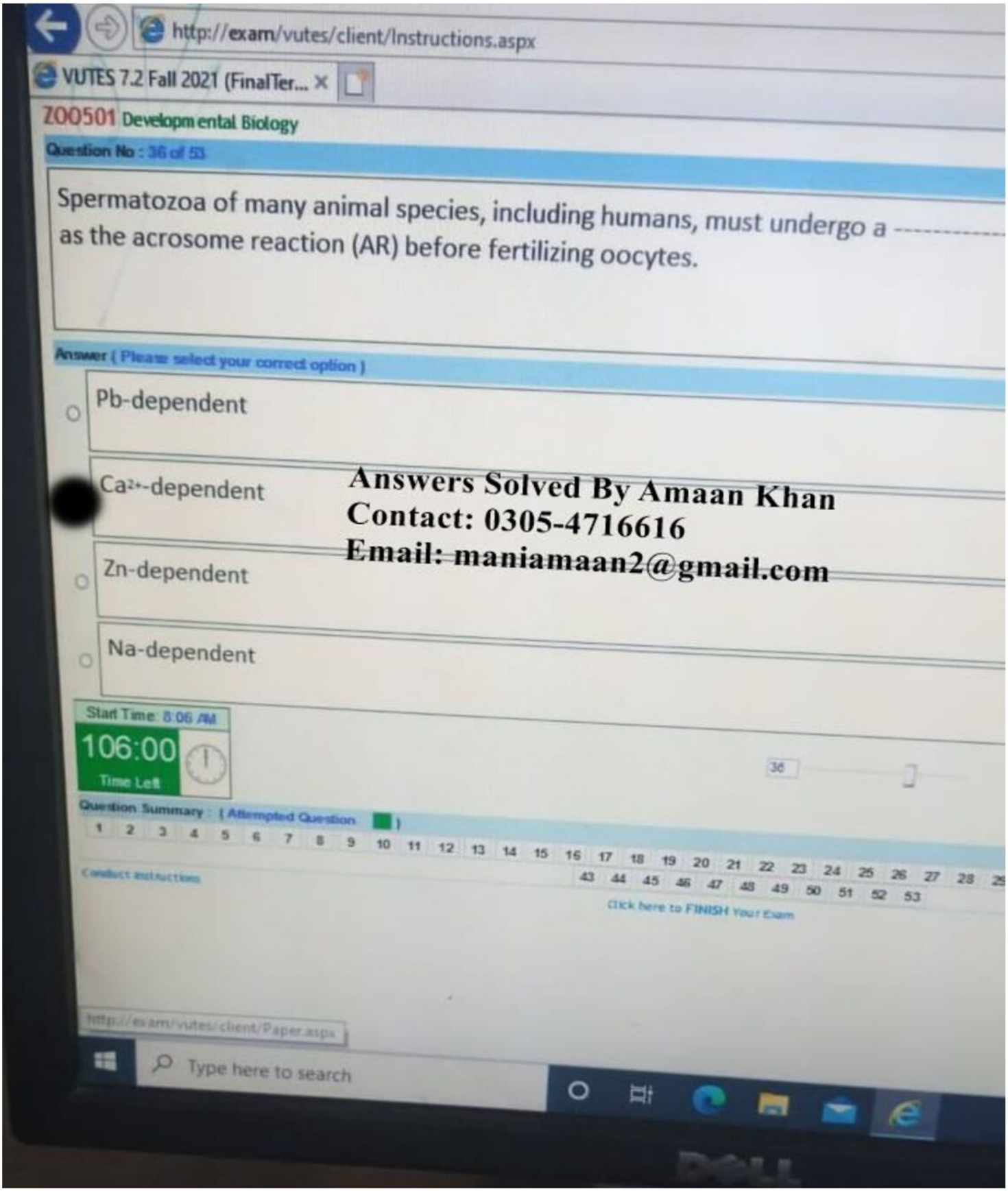
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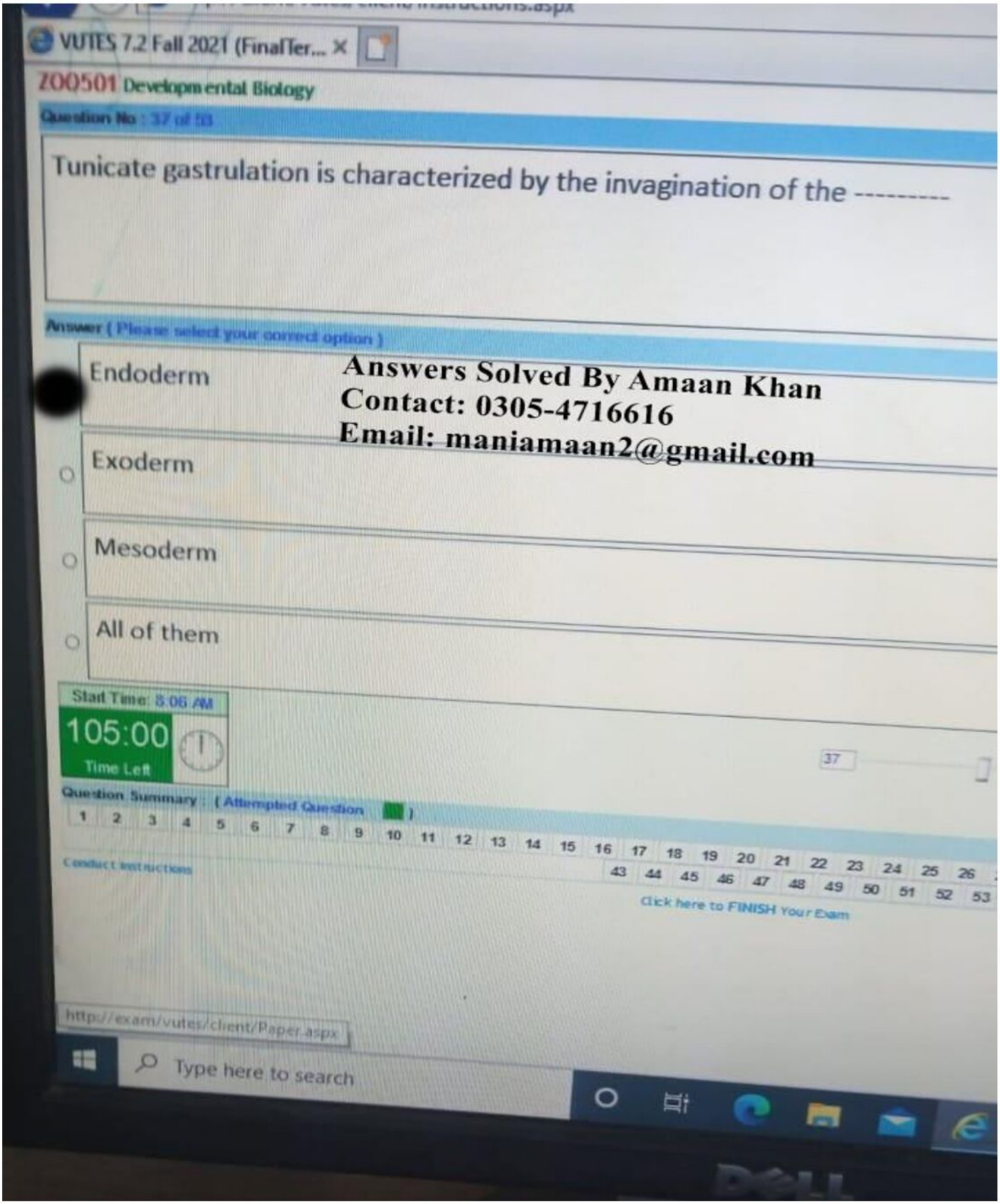
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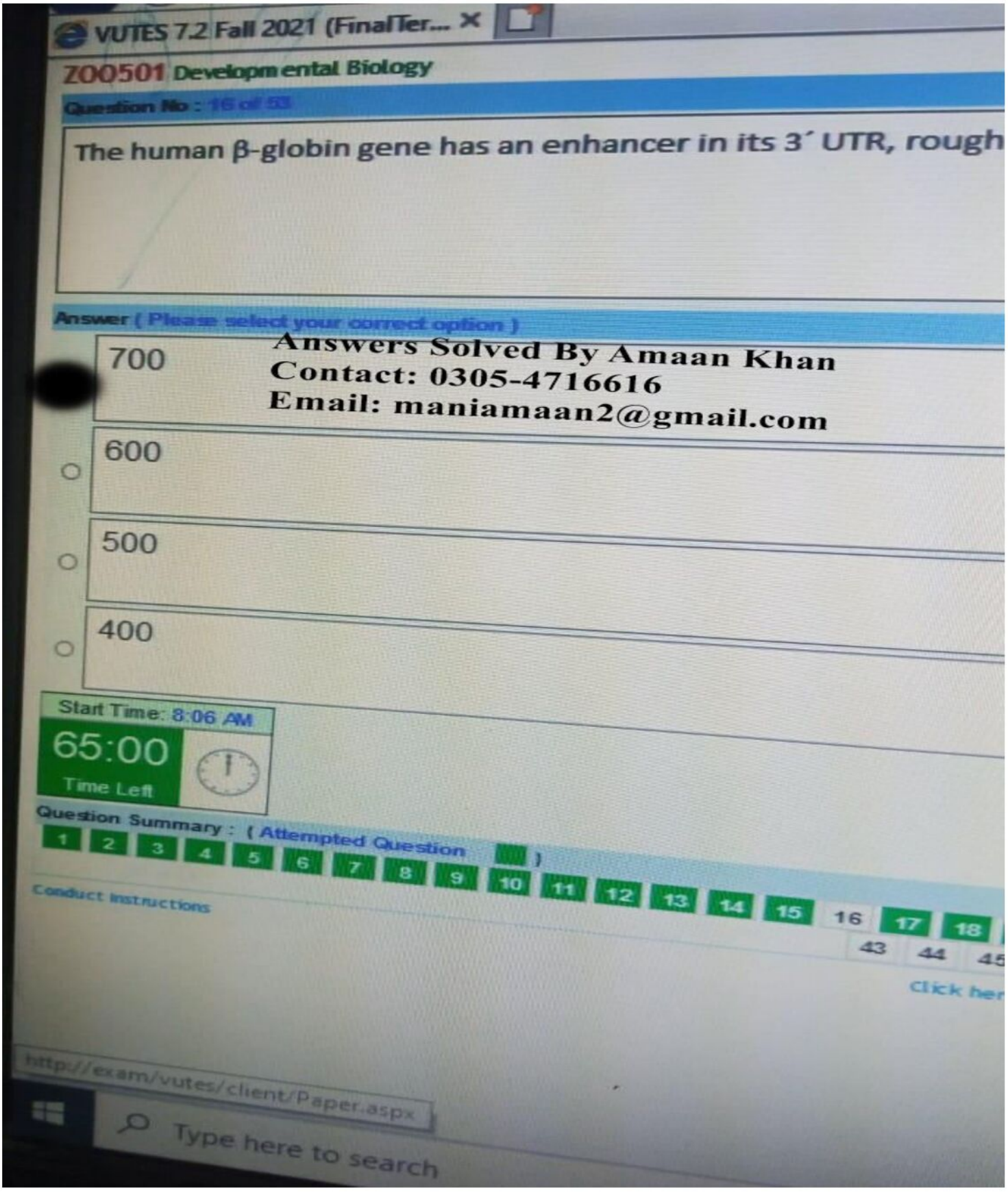
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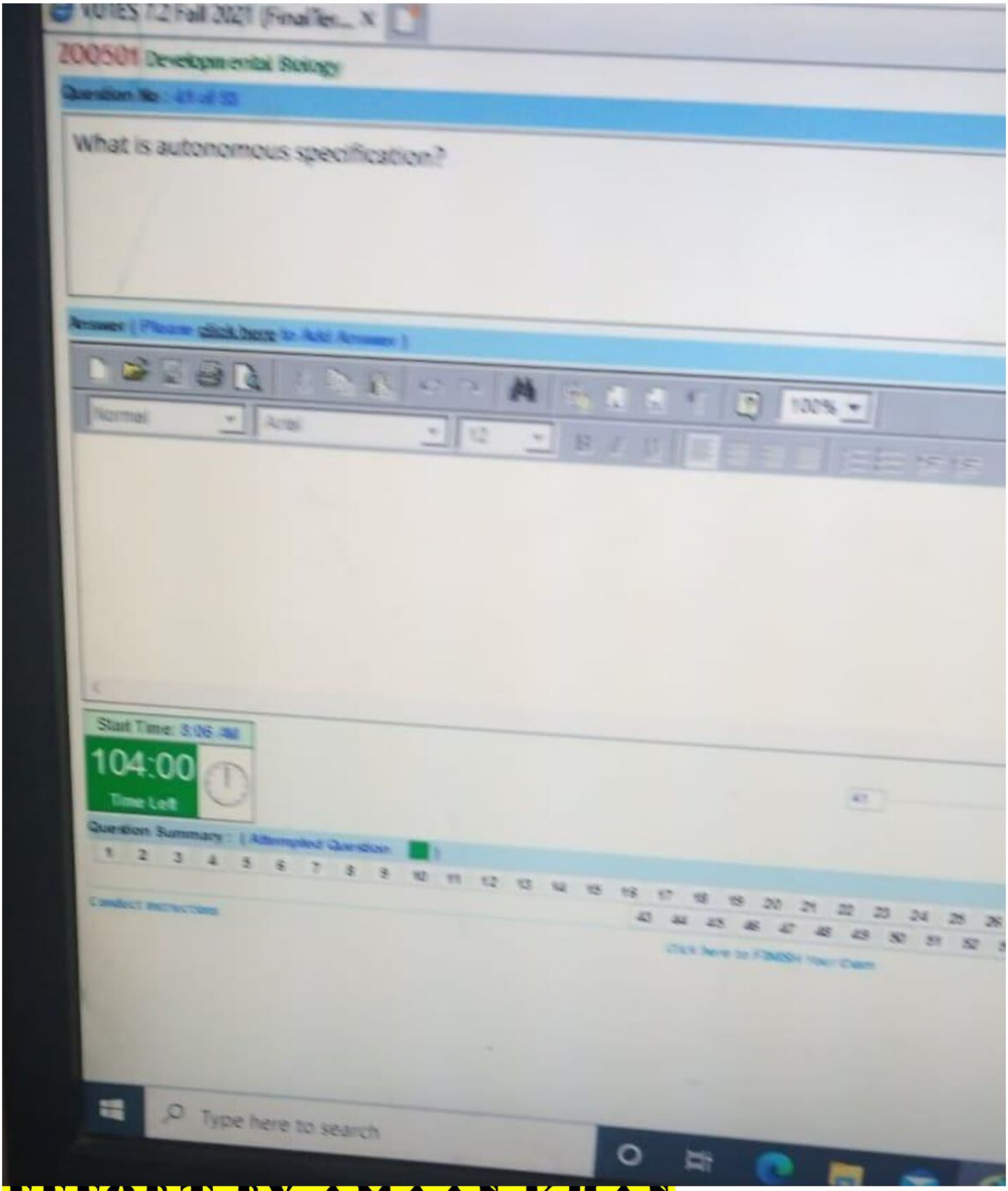
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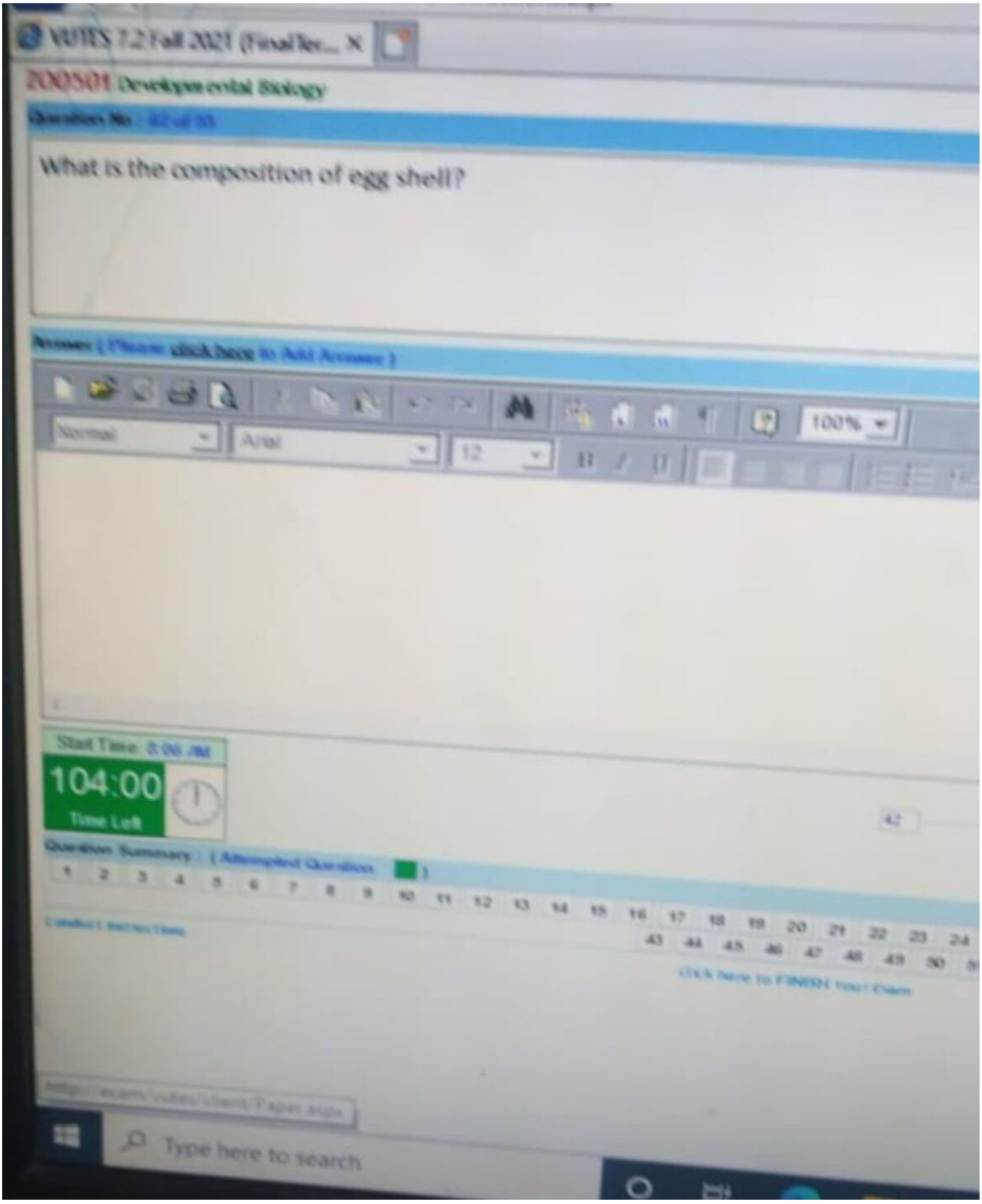
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Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

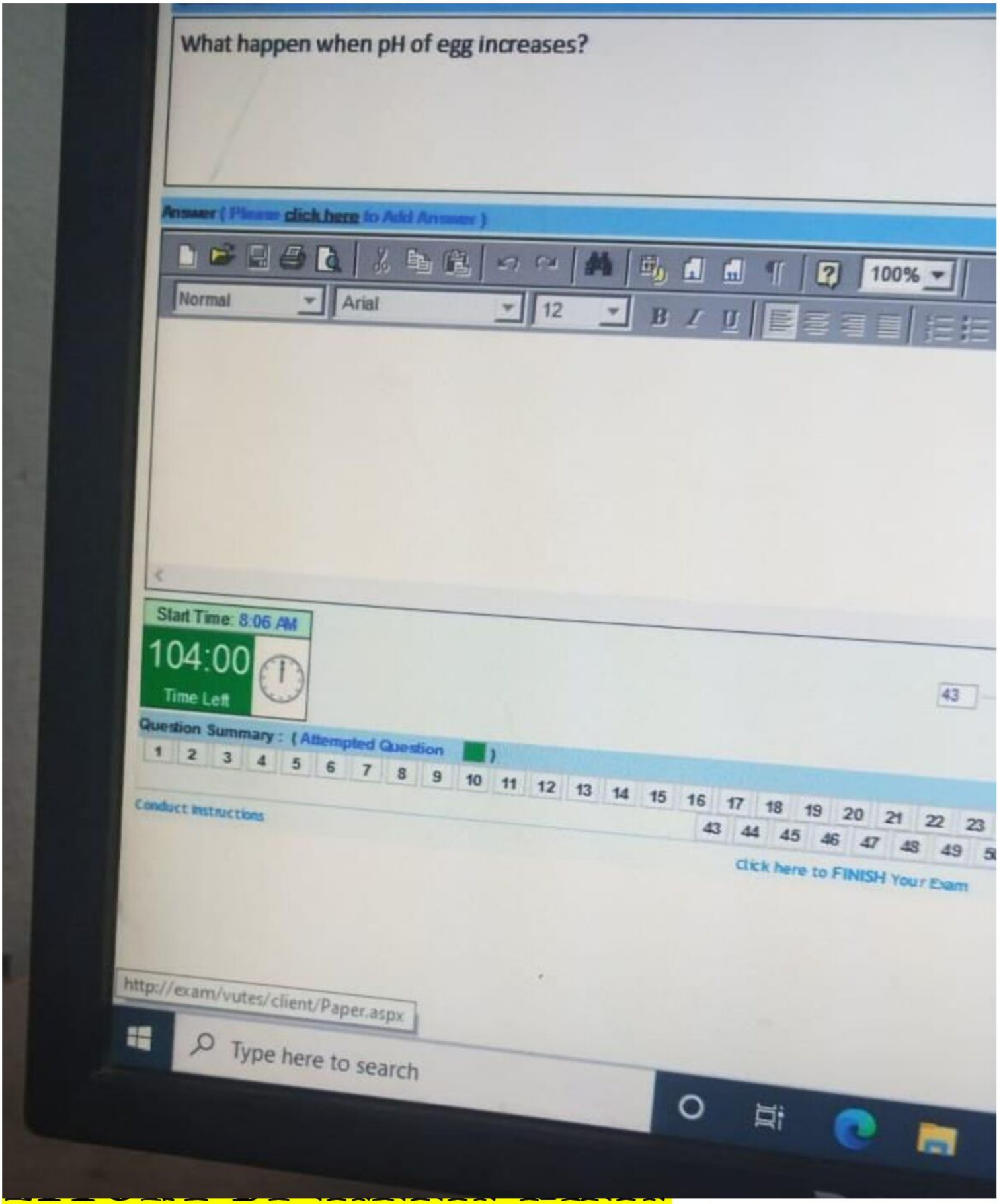
- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.

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Eggshell is made almost entirely of calcium carbonate ($CaCO_3$) crystals. It is a semipermeable membrane, which means that air and moisture can pass through its pores. The shell also has a thin outermost coating called the bloom or cuticle that helps keep out bacteria and dust.

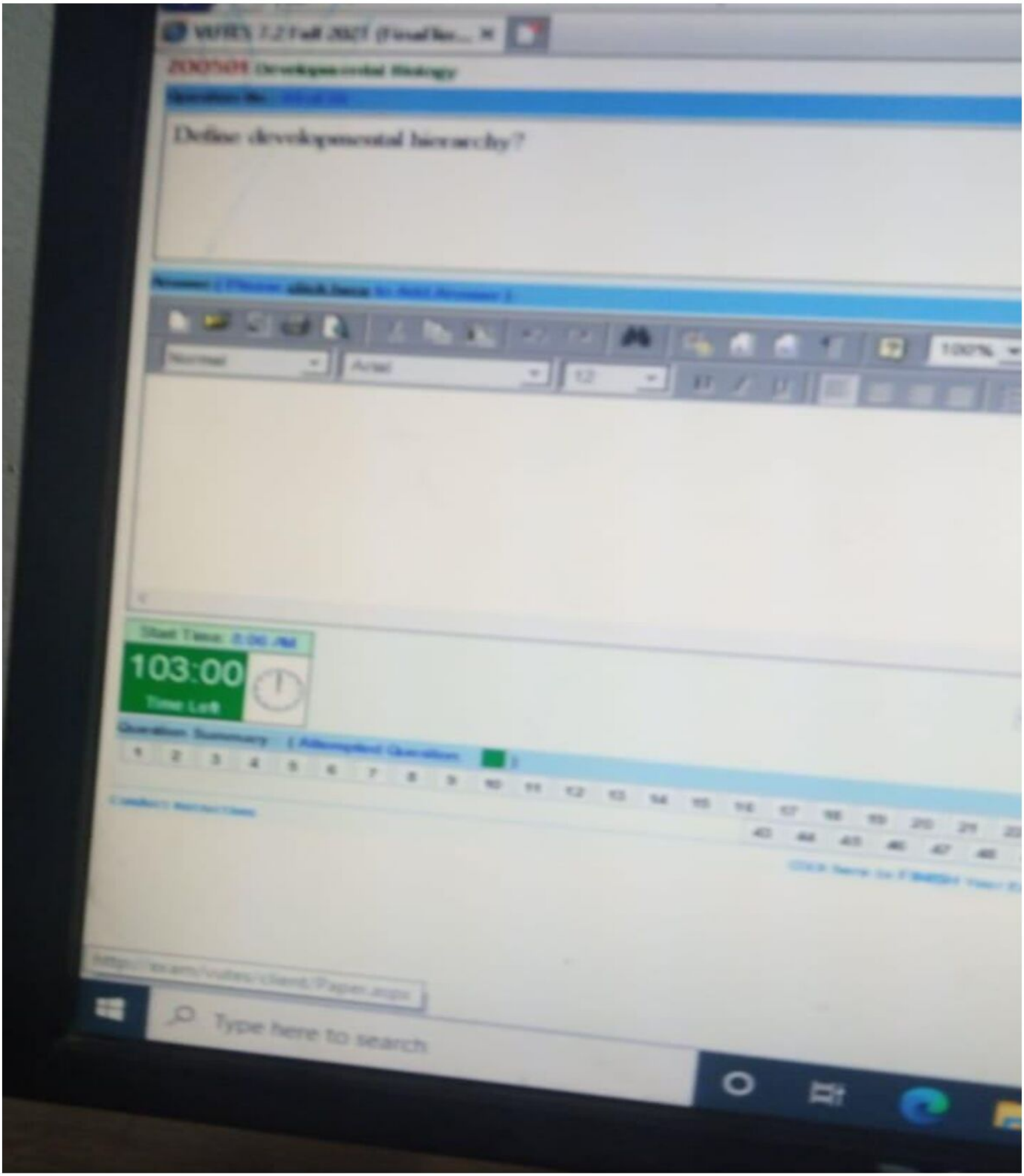
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Late responses:

Shortly after the calcium ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a second influx of sodium ions, which causes a 1:1 exchange between sodium ions from the seawater and hydrogen ions from the egg.

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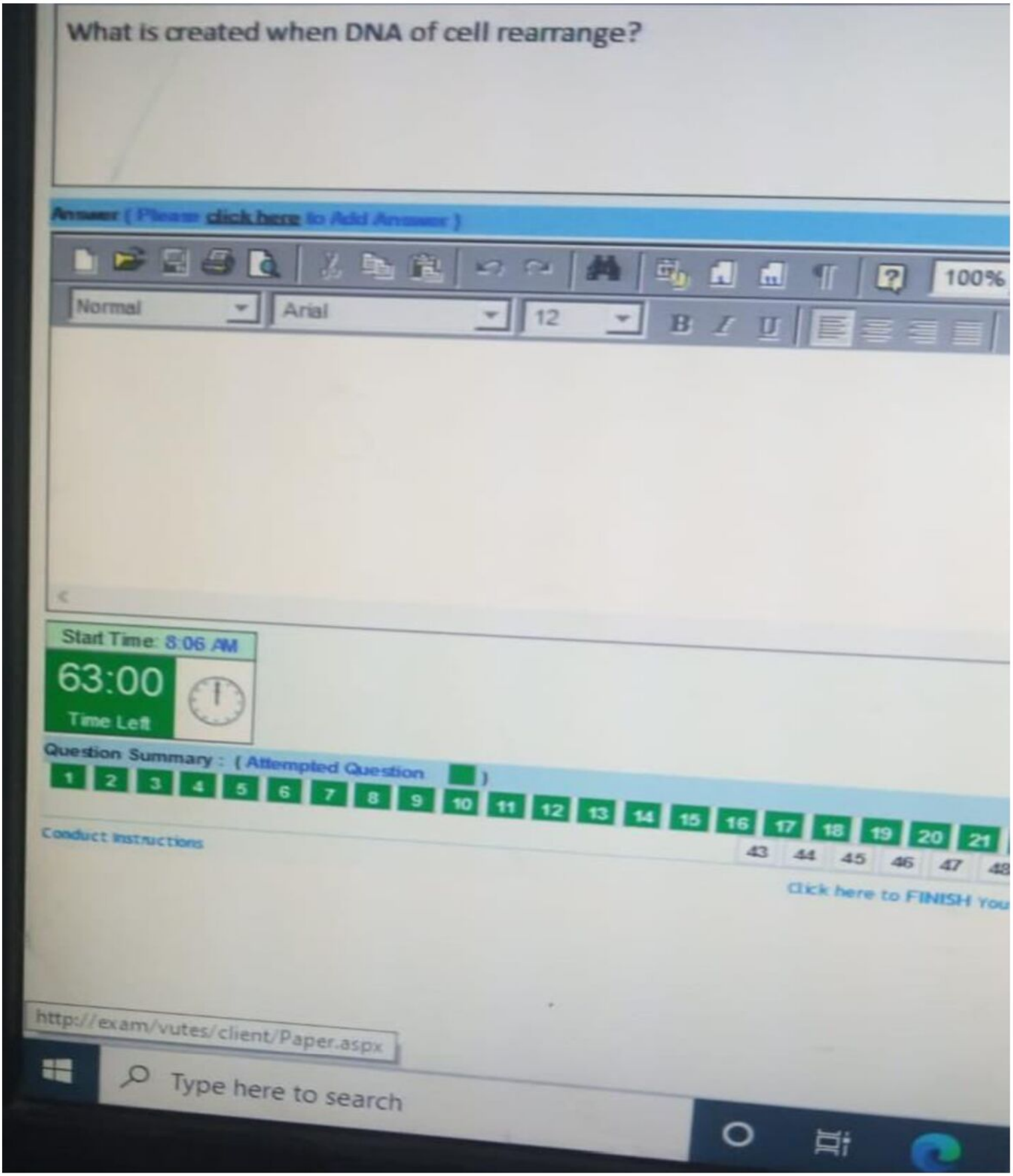


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The **developmental hierarchy** is the hierarchical series of decisions involving specific pre-existing cell types.

These developmental decisions are usually irreversible, that progressively and irreversibly restrict cell fate.

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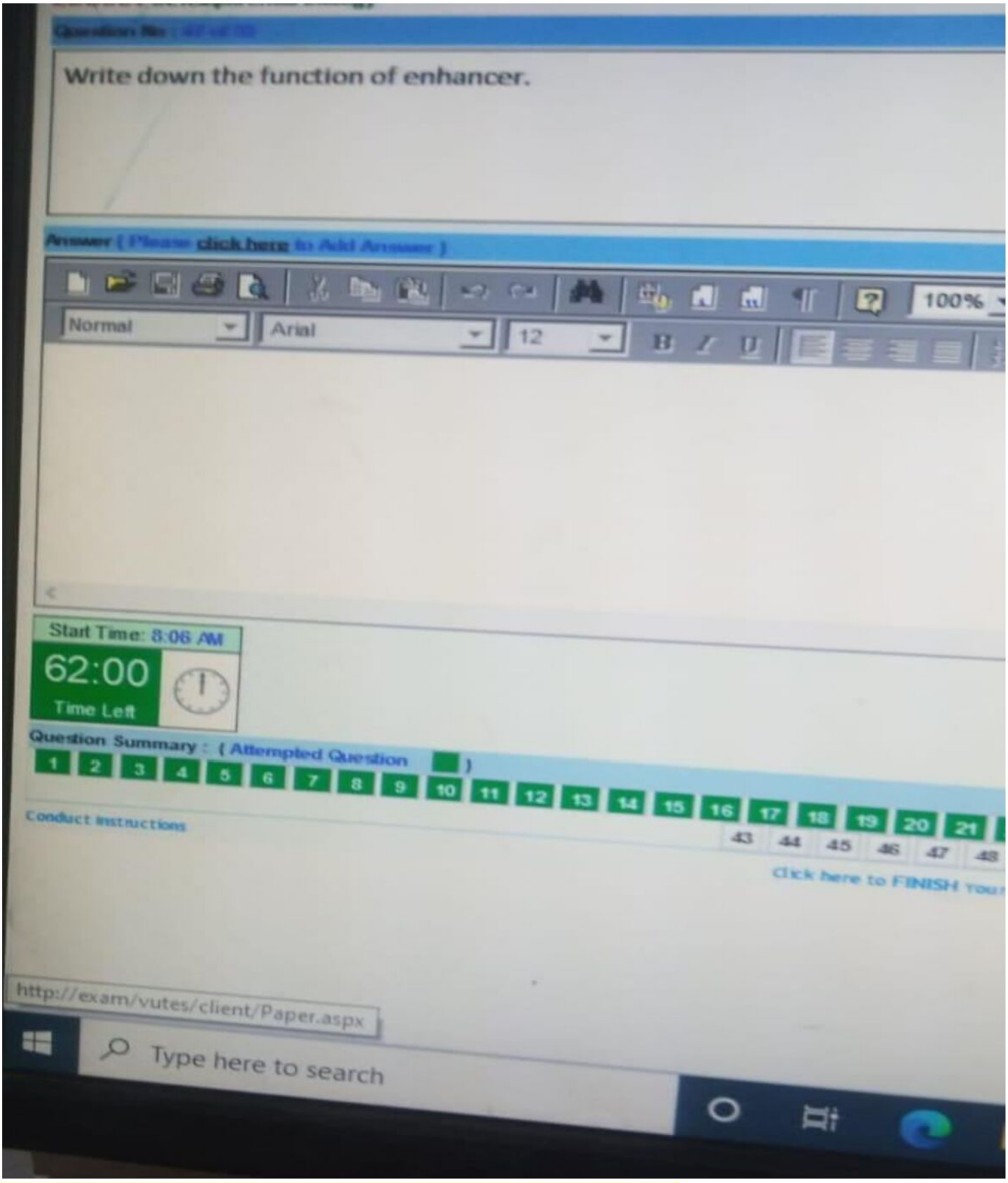


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- The exceptions to the rule of genomic equivalence are the lymphocytes. During differentiation, these cells rearrange their DNA to create new immunoglobulin and antigen receptor genes.

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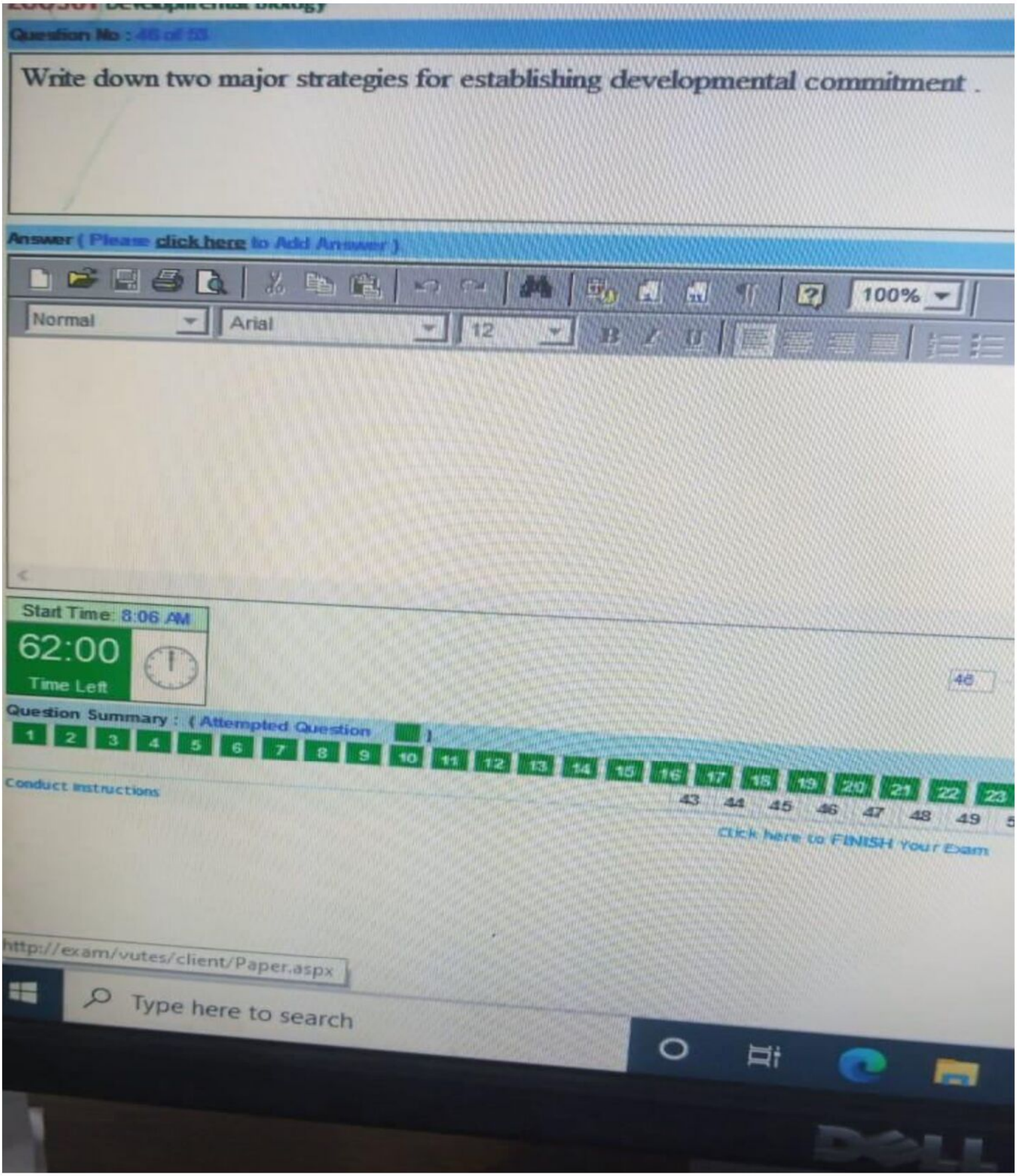
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Enhancer:

It is a DNA sequence that can activate the utilization of a promoter, controlling the efficiency and rate of transcription from that particular promoter.

The human β-globin gene has an enhancer in its 3′ UTR, roughly 700 base pairs.

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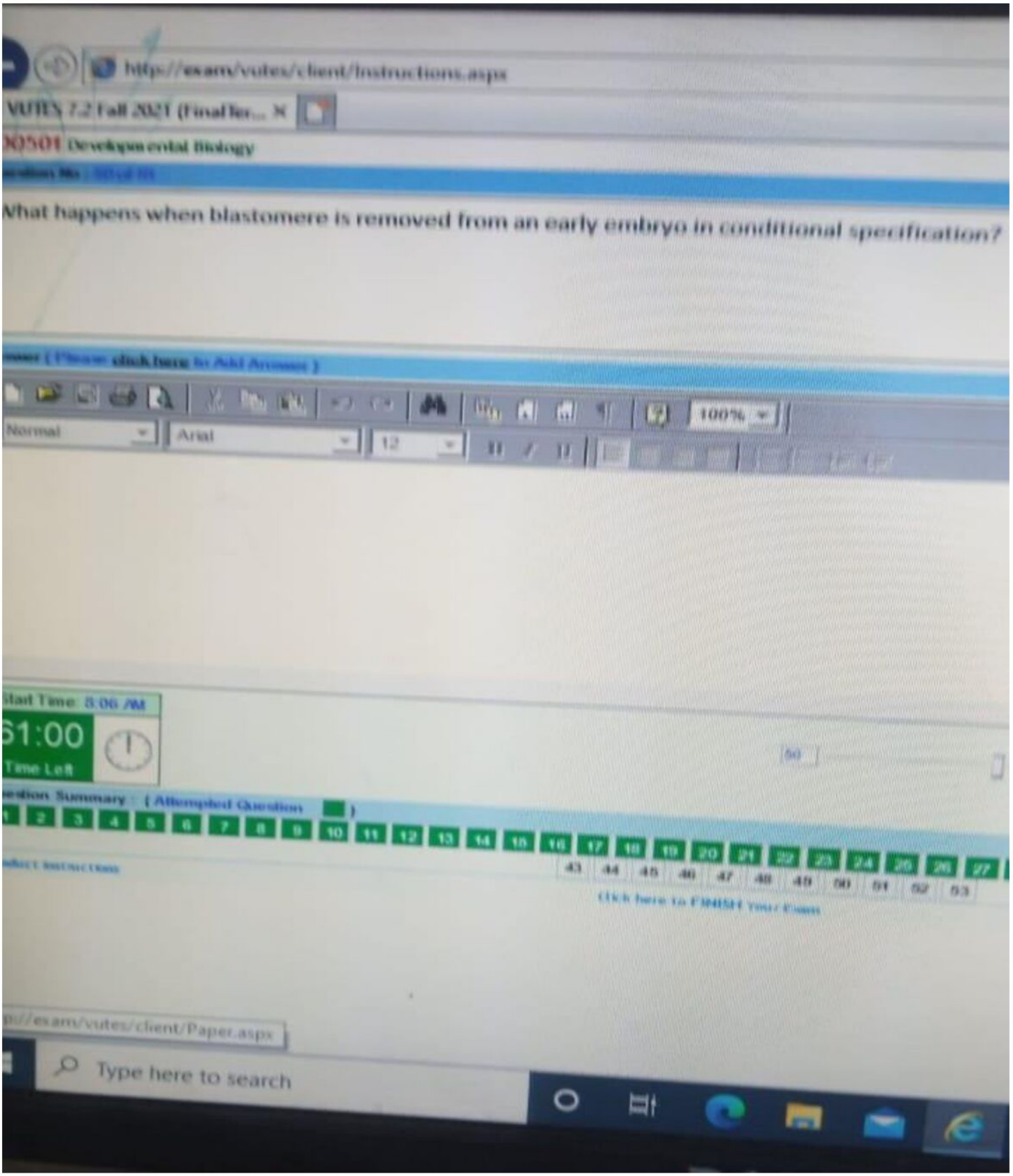


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- **The inheritance of cytoplasmic determinants:** Cytoplasmic determinants are the molecules in cytoplasm that can help to determine cell fate. The asymmetric distribution of cytoplasmic determinants indicates that the mechanism of differentiation is entirely intrinsic.

- **The perception of external inductive signals:** The process where one cell or group of cells changes developmental fate of another is termed induction. It is extrinsic process that depends on the position of a cell in the embryo.

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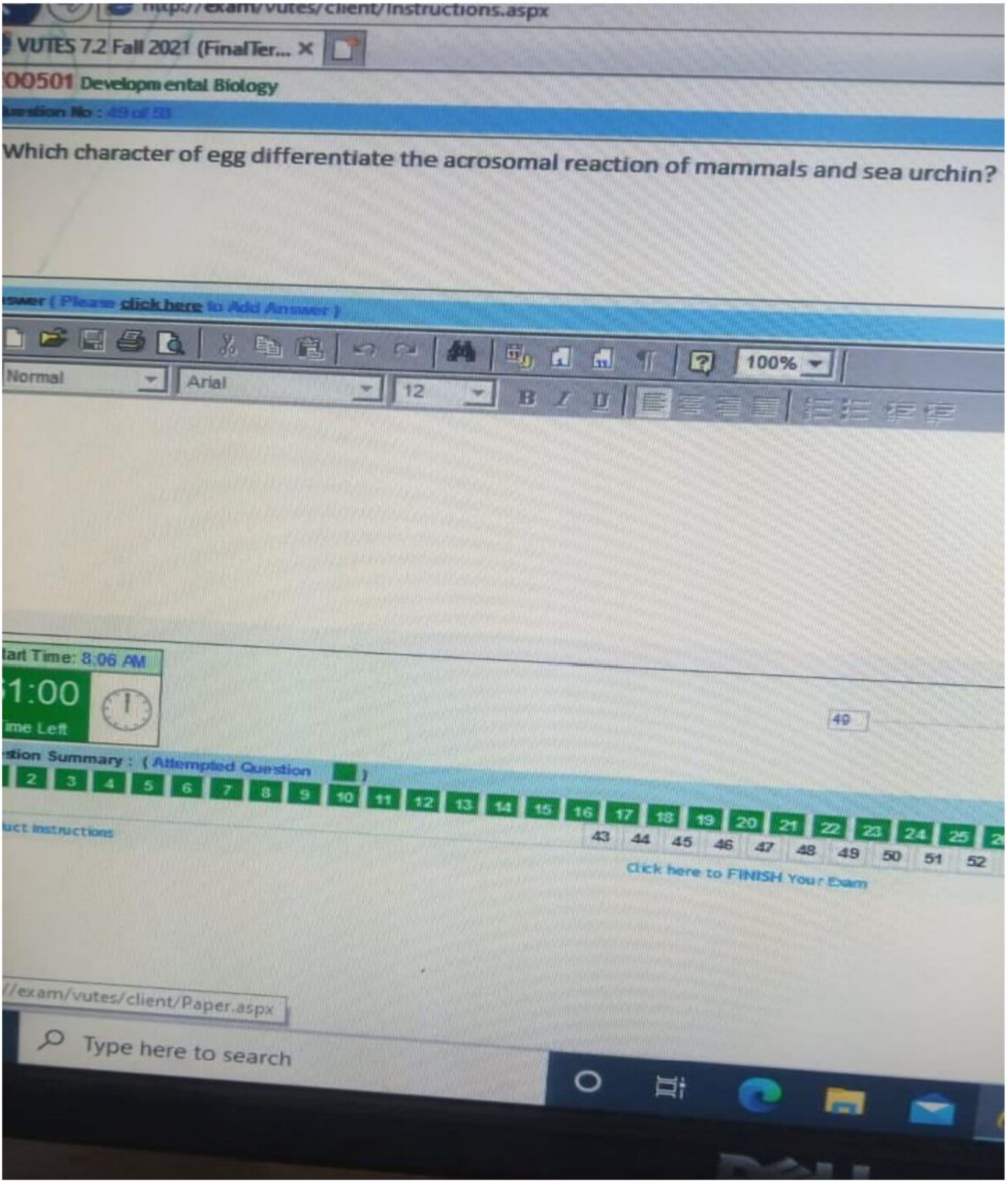


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If a blastomere is removed from an early embryo that uses conditional specification, the remaining embryonic cells alter their fates so that the roles of the missing cells can be taken over.

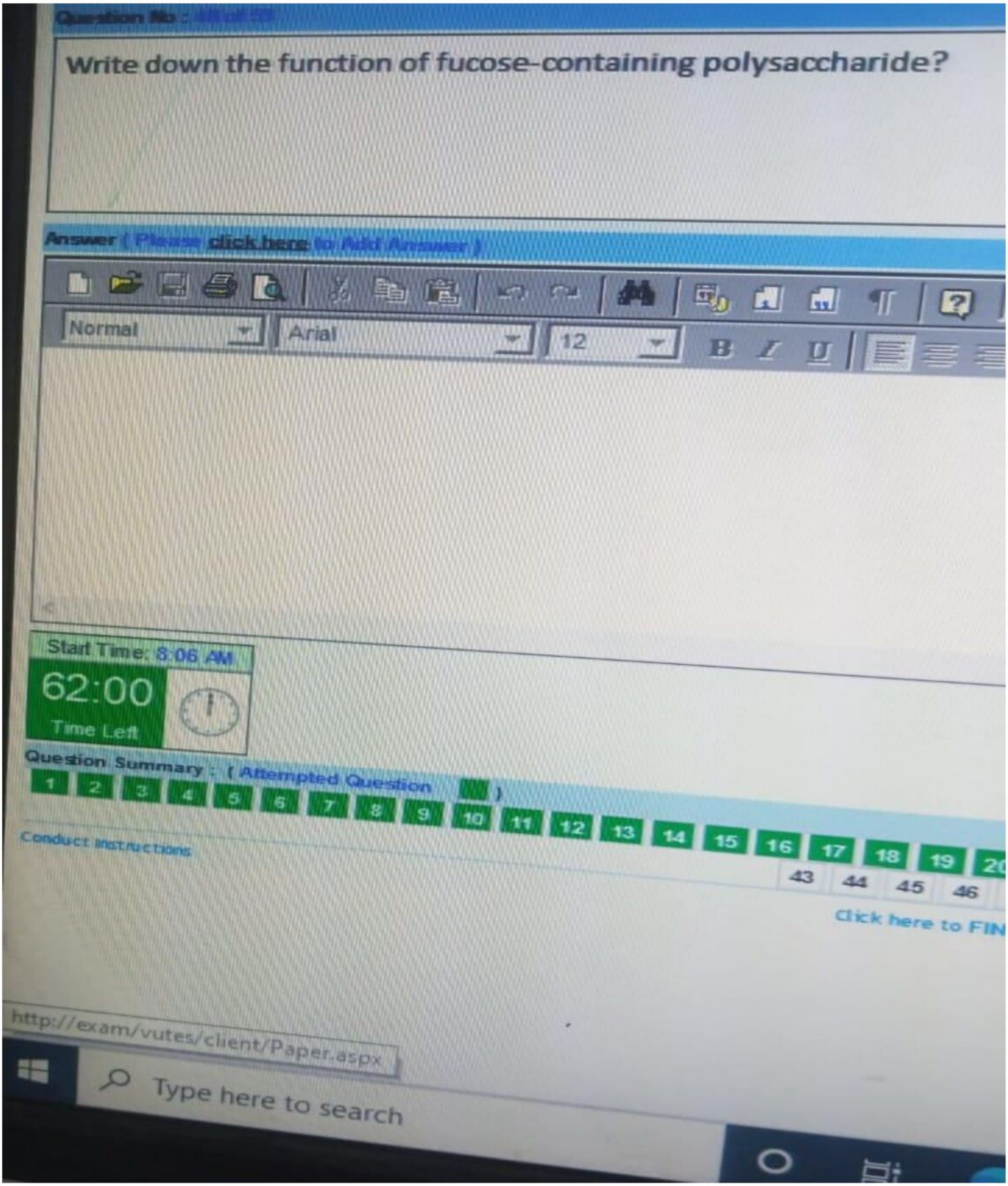
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The acrosomal reaction in sea urchins

In sea urchins, the acrosomal reaction is thought to be initiated by a fucose-containing polysaccharide in the egg jelly that binds to the sperm and allows calcium to enter into the sperm head.

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